PARIS
LODRON
UNIVERSITÄT
SALZBURG

FINAL PROJECT REPORT

DATABASE BACKEND FOR WEBGIS

CHRISTMAS MARKET IN GDAŃSK, POLAND

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Spatial Databases

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1 INTRODUCTION

A web-based platform or mobile application that integrates GIS capabilities offers a unique opportunity for event organizers to provide interactive and dynamic spatial services to users. For this database, the backend plays an important role in efficient spatial data management and queries.

The main objective of the project is to develop a spatial database that serves as a backend for a city festival to fulfill the following requirements:

- Organise the spaces/stalls/attractions as well as other facilities.
- User-specific, dynamic queries that return which events take place or facilities that are close to the current visitor's position.
- User-specific and dynamic queries about the festival, such as digital maps, lists of events etc.

For the case study, the Christmas fair in Gdańsk was selected. It is a popular Christmas town that attracts crowds of interested people every year. Visitors can buy Christmas decorations, original gifts, Christmas delicacies, play games, enjoy events and performances, and try cuisines from all over the world.

2 STUDY AREA

Figure 1: Christmas Market in Gdansk, Source: (The magic of Christmas in the center of Gdansk. Crowds at the Christmas Market, 2024)

Figure 2: Map of the Christmas Market, Gdansk, Source: (The magic of Christmas in the center of Gdansk. Crowds at the Christmas Market, 2024)

3 DATA PREPARATION

3.1 Georeferencing

Using the georeferencer tool in QGIS, map of the christmas market is georeferenced with the OSM base map. Four ground control points were selected and added to the tool. Figure 3 represents the georeferencer tool with the selected GCP (Ground Control Points) enclosed with a purple box. Figure 4 is the result of georeferencing.

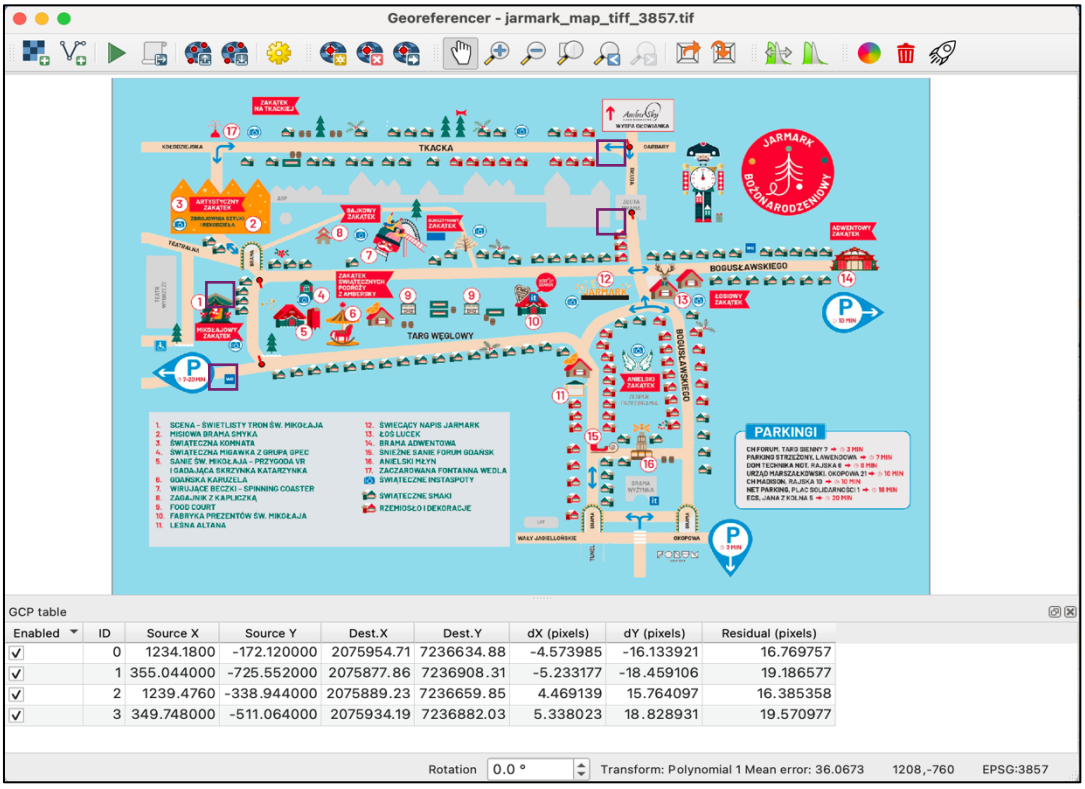


Figure 3: Georeferencing

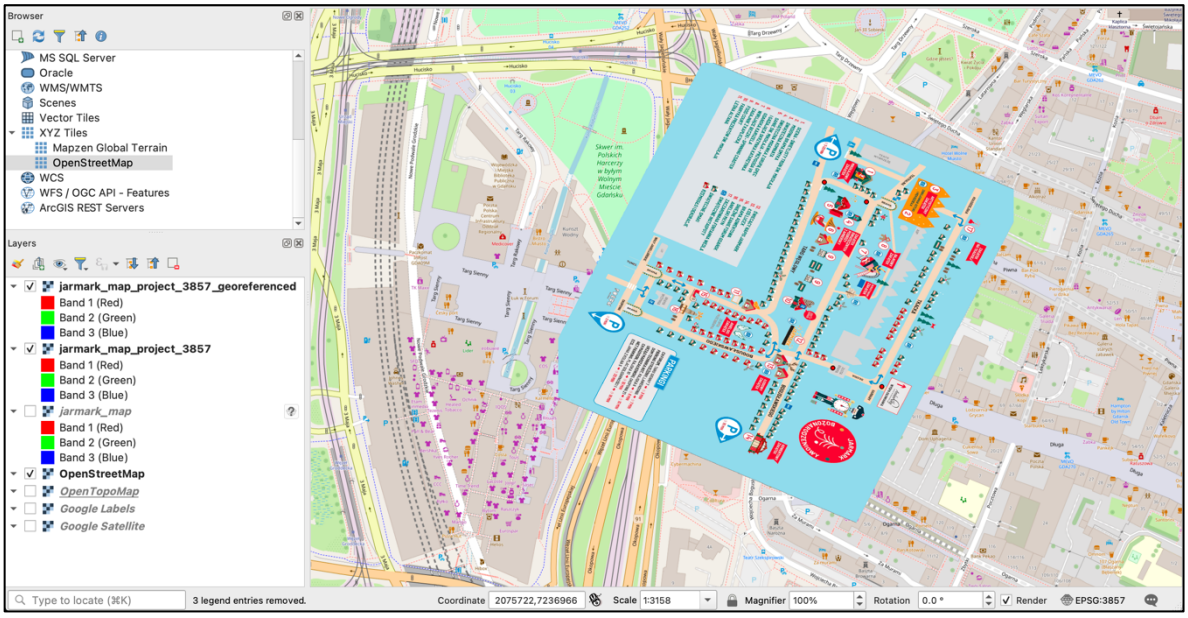
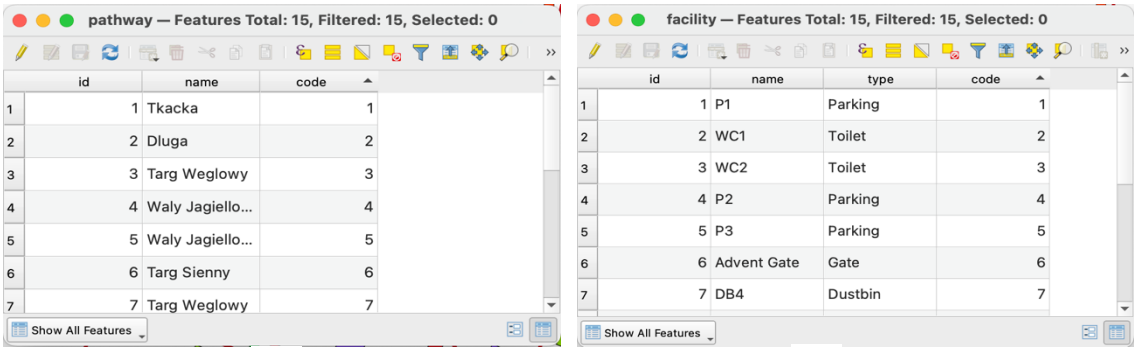


Figure 4: Georeferenced Christmas Market Map

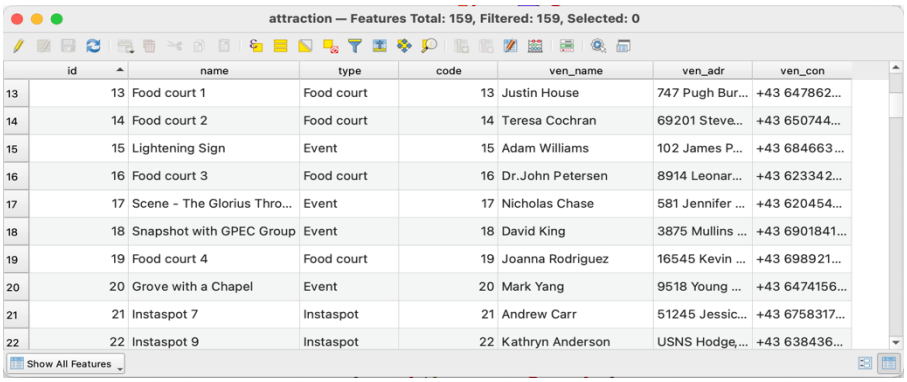
3.2 Digitization and data population

Three new shapefile layers were created to digitise and store information related to facility, pathway, and attractions of point, linestring, and polygon geometry type, respectively, in the EPSG:3857 coordinate system. Figure 5 represents the digitized map and Figures 6 (a), (b), and (c) represent the attribute table of pathway, facility, and attraction layer, respectively.

Figure 5: Digitized Map



a b



c

Figure 6: Attribute Tables, [1]Pathway [b] Facility [c] Attraction

4 IMPLEMENTATION

4.1 Create Database and Create PostGIS Extension

New database was created by running following SQL command in the default postgres query tool.

```
CREATE DATABASE city_festival_management;
```

Then, the respective query tool was opened for the newly created database and the command shown in Figure 7 was runned for creating the PostGIS extension.

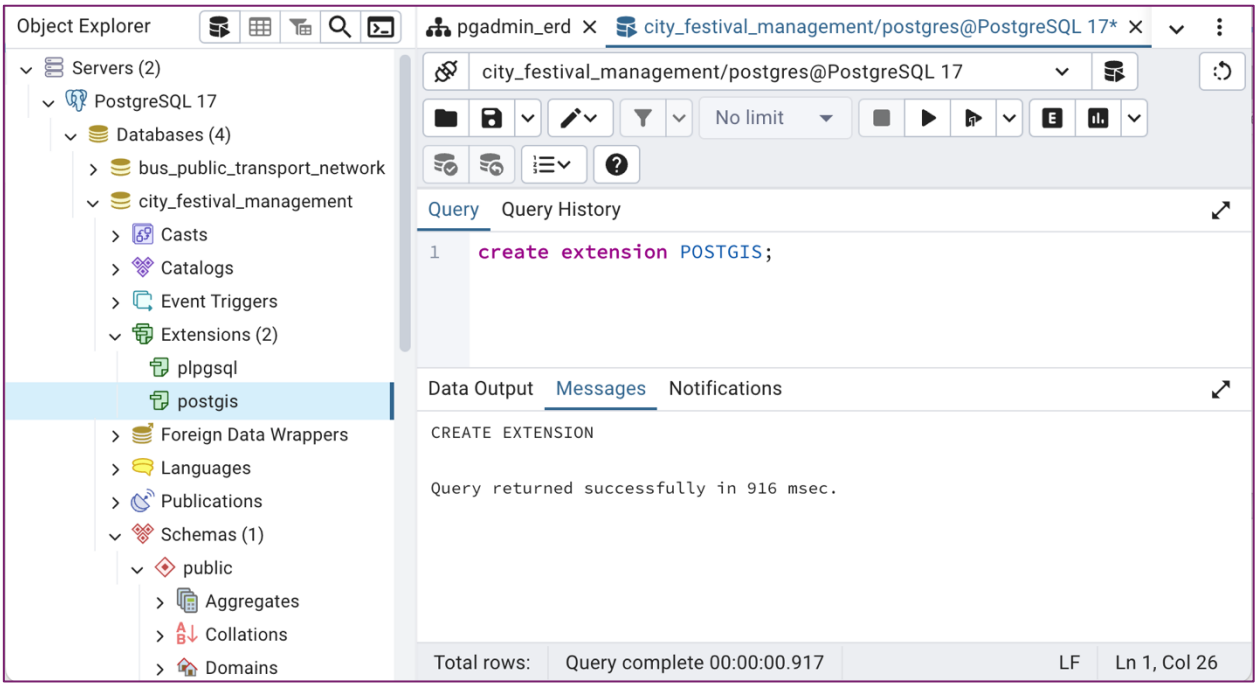


Figure 7: PostGIS Extension Creation

4.2 Connect Database in QGIS

In QGIS, a new PostGIS connection was established for the newly created database, as shown in Figure 8.

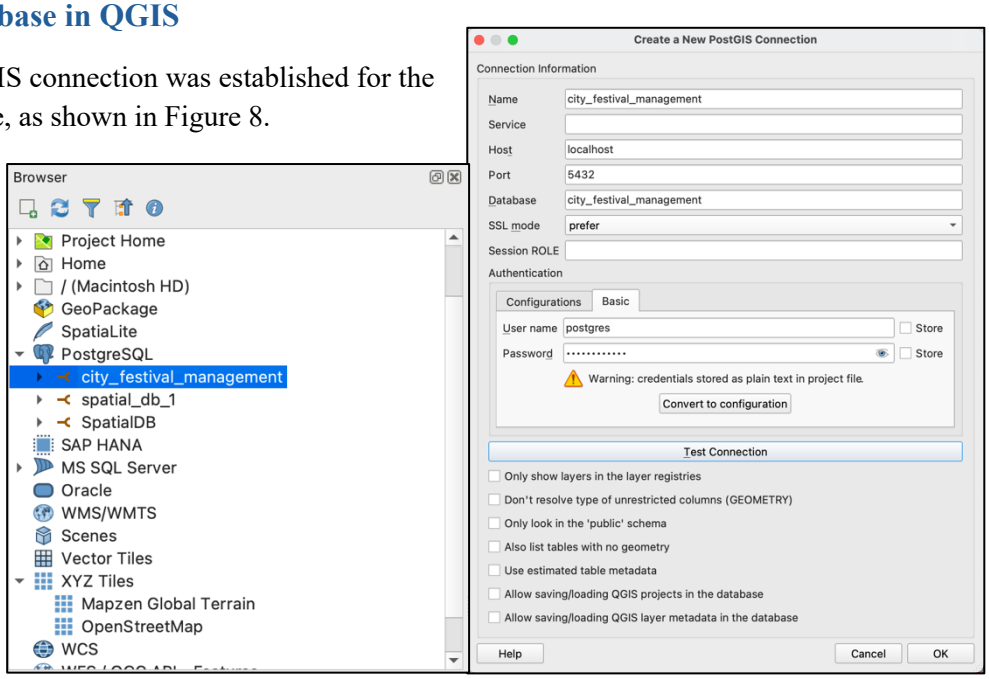


Figure 8: PostGIS Connection in QGIS

4.3 Load Shapefiles into Database

Facility, pathway, and attraction shapefiles were loaded in the database in public schema using DB Manager in QGIS with name 'facility_raw', 'pathway_raw', and 'attraction_raw', respectively as shown in Figure 9.

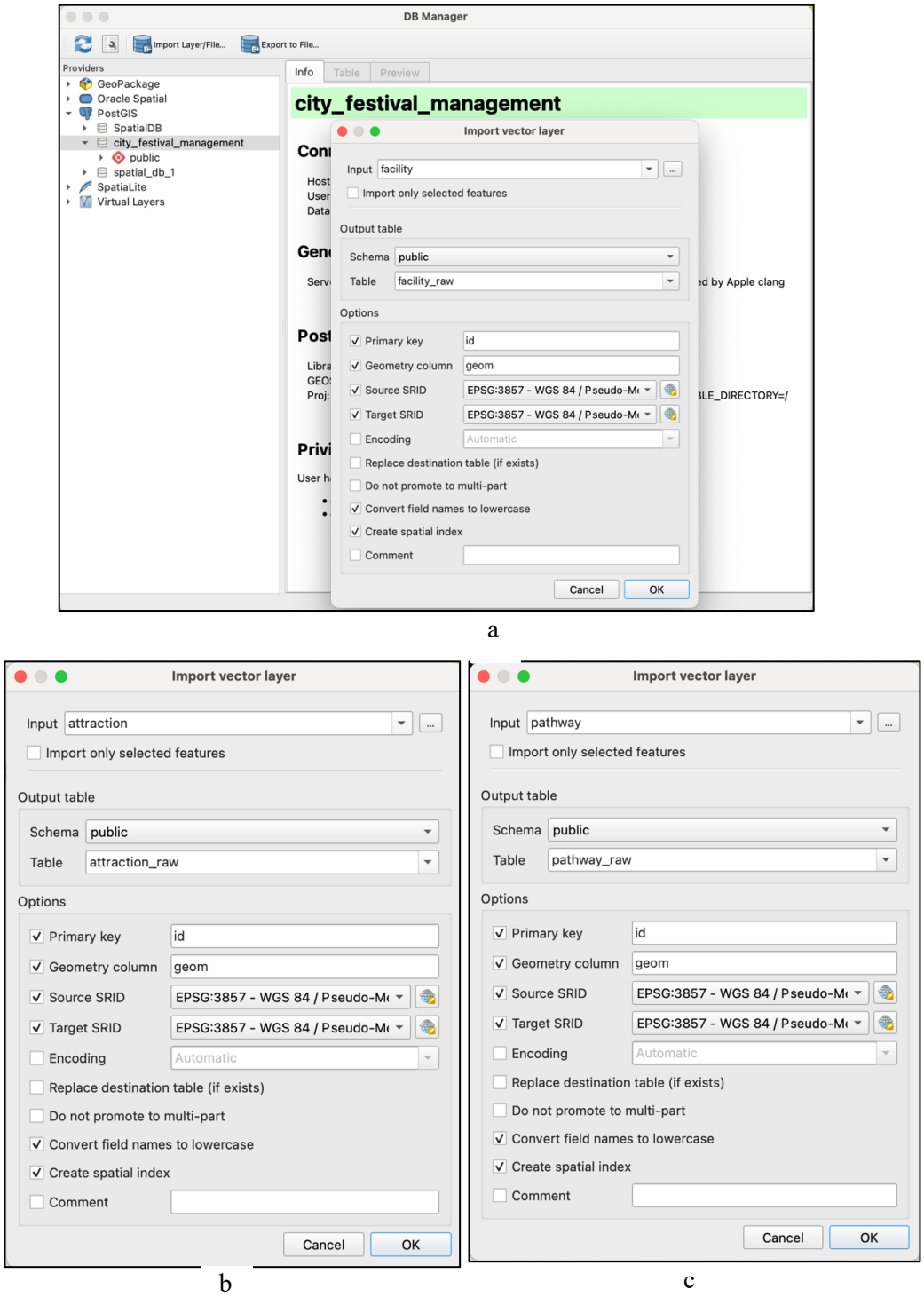


Figure 9: Import Layers into Database

4.4 Build Entity-Relation Diagram using ERD Tool in Pgadmin4

Using ERD tool in Pgadmin, database was designed. Database tables were defined with columns specified by their data types (e.g., integer, varchar, date, geometry), and and properties such as NOT NULL, and DEFAULT values. In addition to this primary key, unique contratints and foreign key relationships were also established. Figure 10 represents the table definition for **attraction**.

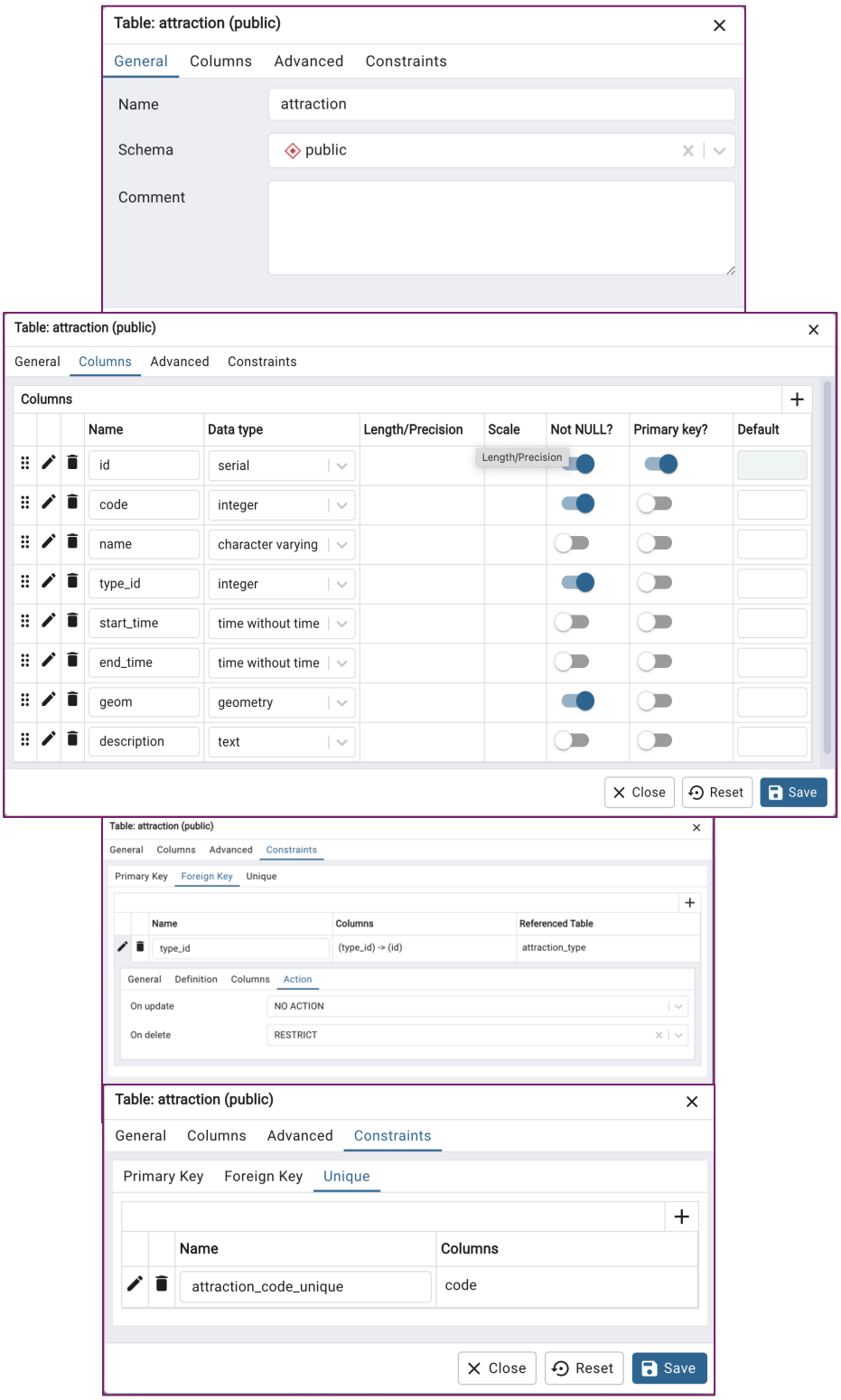


Figure 10: Table Definition in ERD Tool

Figure 11 represents the ER diagram for the city festival management database. The facility table would stores all kinds of facilities available such as toilet, water station, parking etc. Similarly, the attraction table is designed to store all kinds of attractions that includes, including food, gift shop, entertainment, events, etc.. The user table is defined with columns name, address, contact, type_id, and description. Here, contacts is defined with constraints unique contact, which serves as a unique identifier to the contacts and is also used for creating roles. The database is structured to fulfill the third normal form (3NF) to eliminate redundancy and ensure data integrity. For instance, user_type, attraction_type, and facility_type tables are created separately only to store types and are linked with the foreign key in corresponding tables. Likewise, considering one vendor might own multiple stalls/attractions, the "attraction_vendor" table is defined separately. Also, the 'facility_staff' table is designed to handle cases where a single staff member might be assigned to multiple facilities.

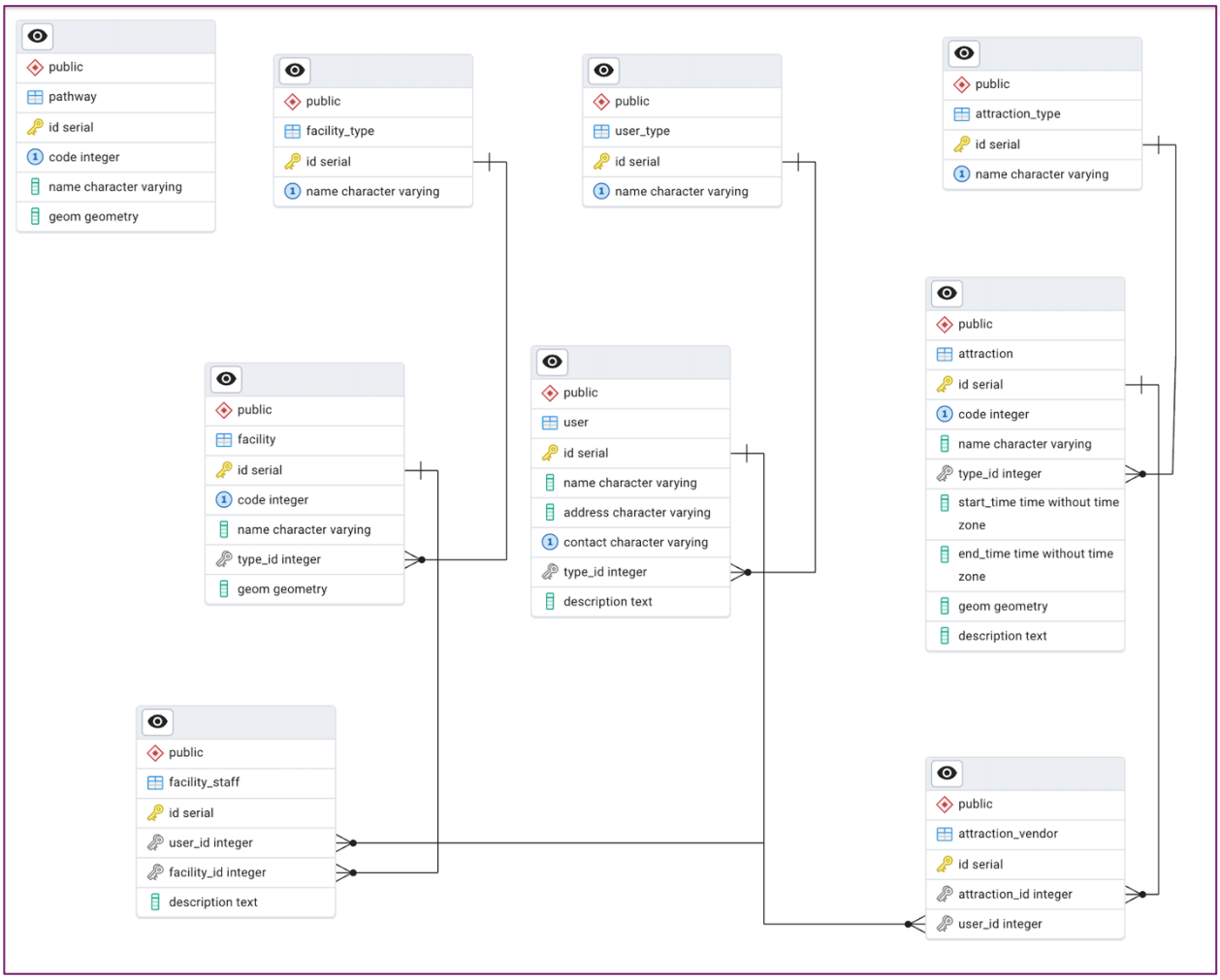


Figure 11: ER Diagram

4.5 Create Tables in Database

After finalizing the ER diagram, SQL was generated as shown below, and SQL was executed to create tables in the database.

```
-- This script was generated by the ERD tool in pgAdmin 4.
-- Please log an issue at https://github.com/pgadmin-
org/pgadmin4/issues/new/choose if you find any bugs, including
reproduction steps.
BEGIN;

CREATE TABLE IF NOT EXISTS public.attraction
(
    id serial NOT NULL,
    code integer NOT NULL,
    name character varying,
    type_id integer NOT NULL,
    start_time time without time zone,
    end_time time without time zone,
    geom geometry NOT NULL,
    description text,
    PRIMARY KEY (id),
    CONSTRAINT attraction_code_unique UNIQUE (code)
);

CREATE TABLE IF NOT EXISTS public.facility
(
    id serial NOT NULL,
    code integer NOT NULL,
    name character varying,
    type_id integer NOT NULL,
    geom geometry NOT NULL,
    PRIMARY KEY (id),
    CONSTRAINT facility_code_unique UNIQUE (code)
);

CREATE TABLE IF NOT EXISTS public.pathway
(
    id serial NOT NULL,
    code integer NOT NULL,
    name character varying,
    geom geometry NOT NULL,
    PRIMARY KEY (id),
    CONSTRAINT pathway_code_unique UNIQUE (code)
);

CREATE TABLE IF NOT EXISTS public."user"
(
    id serial NOT NULL,
    name character varying NOT NULL,
    address character varying,
    contact character varying NOT NULL,
    type_id integer,
    description text,
    PRIMARY KEY (id),
    CONSTRAINT user_contact_unique UNIQUE (contact)
);

CREATE TABLE IF NOT EXISTS public.facility_type
(
    id serial NOT NULL,
    name character varying NOT NULL,
    PRIMARY KEY (id),
    CONSTRAINT facility_name_unique UNIQUE (name)
);
```

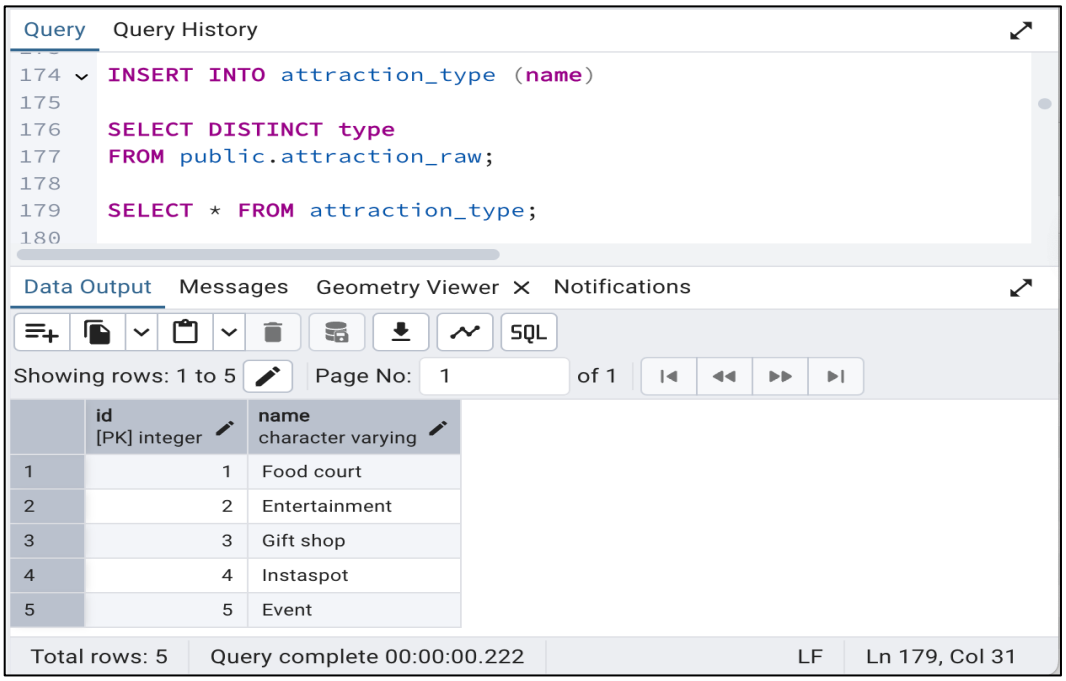
```sql
CREATE TABLE IF NOT EXISTS public.attraction_type
(
    id serial NOT NULL,
    name character varying NOT NULL,
    CONSTRAINT id PRIMARY KEY (id),
    CONSTRAINT attraction_type_name_unique UNIQUE (name)
);

CREATE TABLE IF NOT EXISTS public.attraction_vendor
(
    id serial NOT NULL,
    attraction_id integer NOT NULL,
    user_id integer NOT NULL,
    PRIMARY KEY (id)
);

CREATE TABLE IF NOT EXISTS public.facility_staff
(
    id serial NOT NULL,
    user_id integer NOT NULL,
    facility_id integer NOT NULL,
    description text,
    PRIMARY KEY (id)
);

CREATE TABLE IF NOT EXISTS public.user_type
(
    id serial NOT NULL,
    name character varying NOT NULL,
    PRIMARY KEY (id),
    CONSTRAINT user_type_name_unique UNIQUE (name)
);

ALTER TABLE IF EXISTS public.attraction
    ADD CONSTRAINT type_id FOREIGN KEY (type_id)
    REFERENCES public.attraction_type (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE RESTRICT;


ALTER TABLE IF EXISTS public.facility
    ADD CONSTRAINT type_id FOREIGN KEY (type_id)
    REFERENCES public.facility_type (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE RESTRICT;


ALTER TABLE IF EXISTS public."user"
    ADD CONSTRAINT type_id FOREIGN KEY (type_id)
    REFERENCES public.user_type (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE RESTRICT;


ALTER TABLE IF EXISTS public.attraction_vendor
    ADD CONSTRAINT user_id FOREIGN KEY (user_id)
    REFERENCES public."user" (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE CASCADE;


ALTER TABLE IF EXISTS public.attraction_vendor
    ADD CONSTRAINT attraction_id FOREIGN KEY (attraction_id)
    REFERENCES public.attraction (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE CASCADE;
```

```
ALTER TABLE IF EXISTS public.attraction_vendor
    ADD CONSTRAINT attraction_id FOREIGN KEY (attraction_id)
    REFERENCES public.attraction (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE CASCADE;


ALTER TABLE IF EXISTS public.facility_staff
    ADD CONSTRAINT user_id FOREIGN KEY (user_id)
    REFERENCES public."user" (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE CASCADE;


ALTER TABLE IF EXISTS public.facility_staff
    ADD CONSTRAINT facility_id FOREIGN KEY (facility_id)
    REFERENCES public.facility (id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE NO ACTION;

END;
```

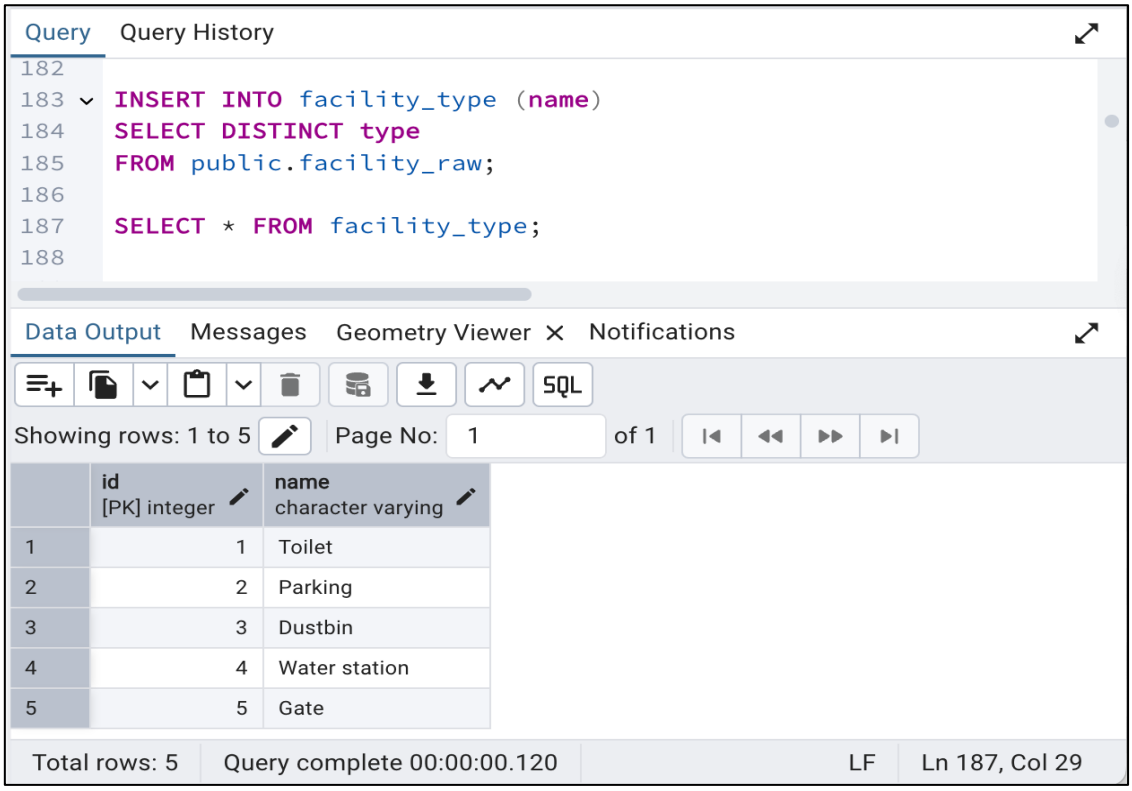
4.6 Insert Data into Database and Database Normalization

In this step, imported raw layers i.e. pathway_raw, facility_raw and attraction_raw, are inserted into the newly created tables.

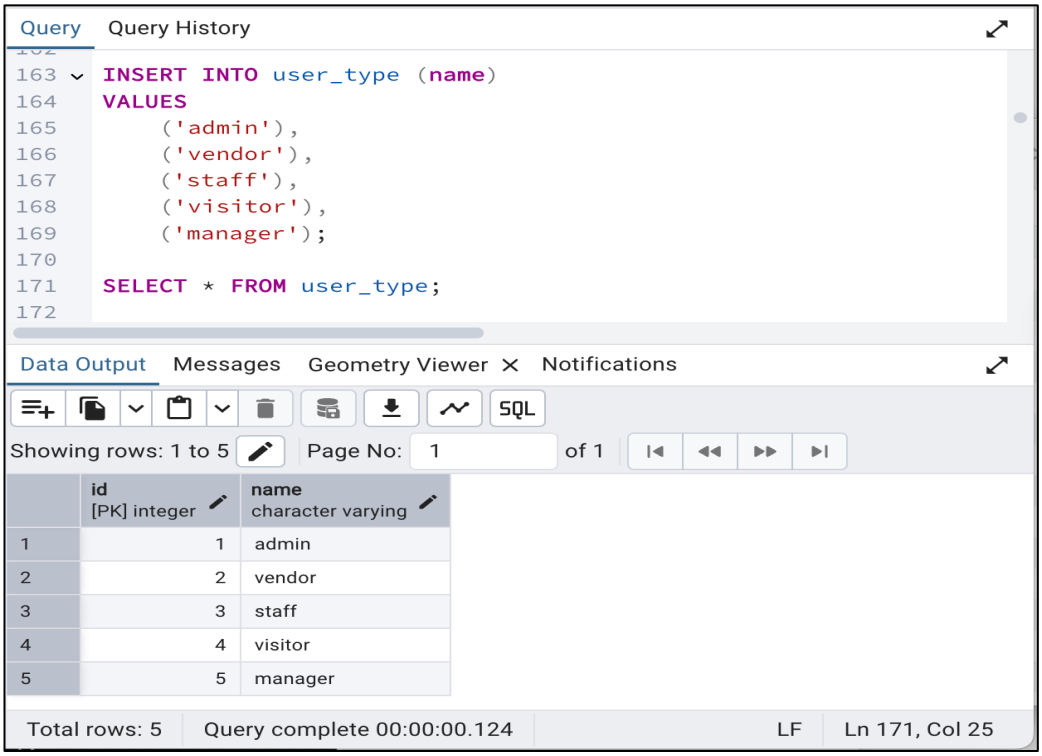
Distinct attraction types are retrieved from attraction_raw table and added into the attraction_type table.



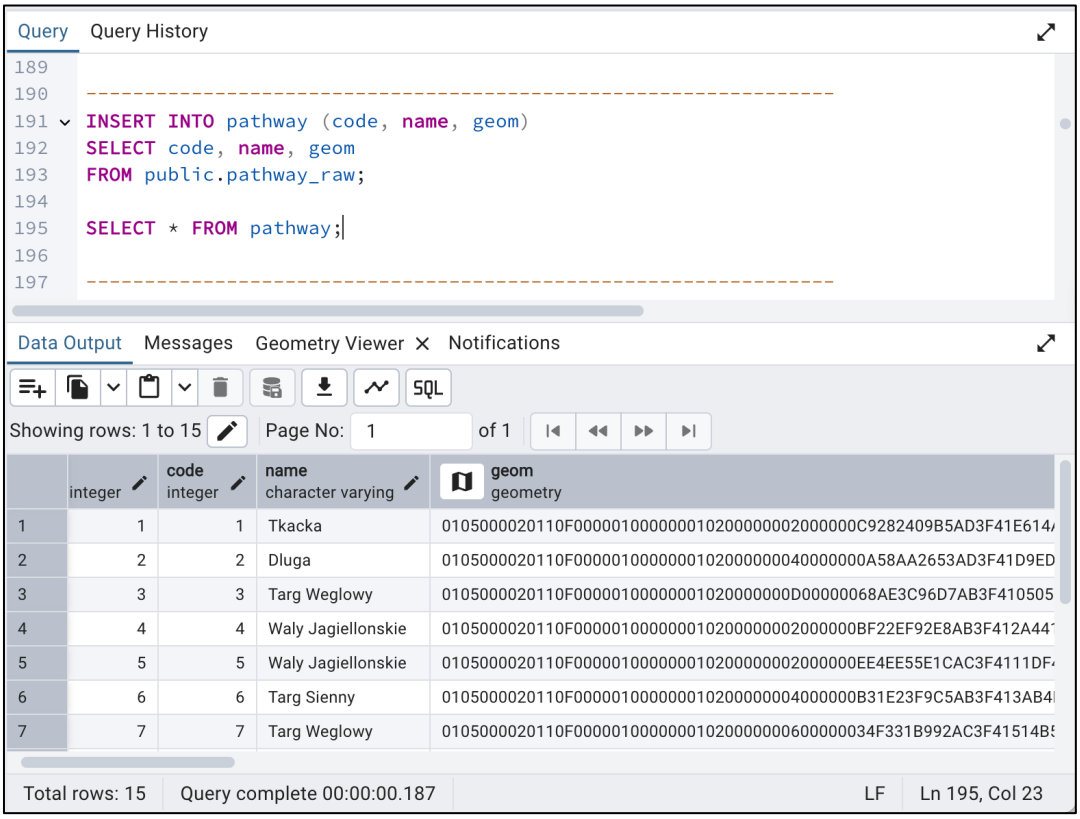
Distinct facility types are retrieved form facility_raw table and added into the facility_type table.



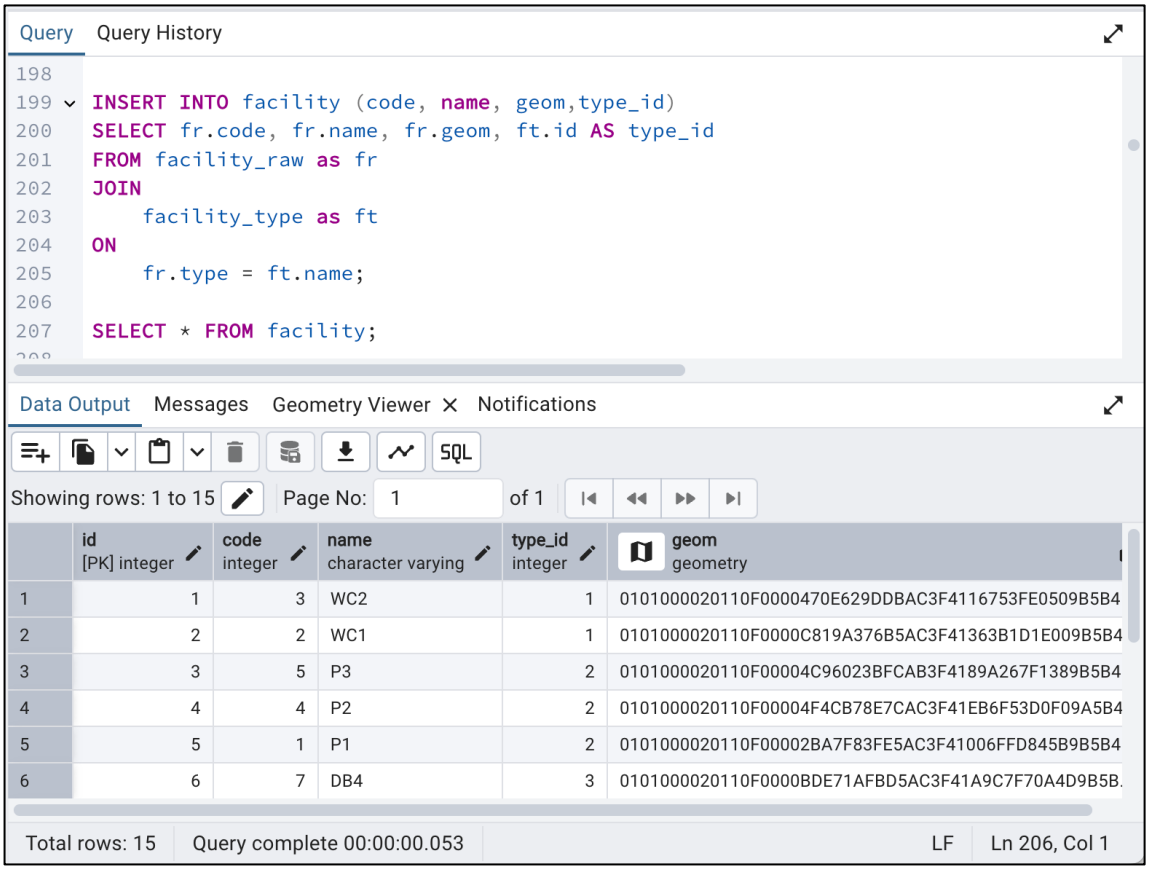
Four user types were inserted into to the user_type table



Pathway table is populated using the pathway_raw data.



Facility table is populated using the facility_raw data and type_id (FK) is extracted from facility_type table.



```
Query  Query History

198
199  INSERT INTO facility (code, name, geom,type_id)
200  SELECT fr.code, fr.name, fr.geom, ft.id AS type_id
201  FROM facility_raw as fr
202  JOIN
203      facility_type as ft
204  ON
205      fr.type = ft.name;
206
207  SELECT * FROM facility;
```

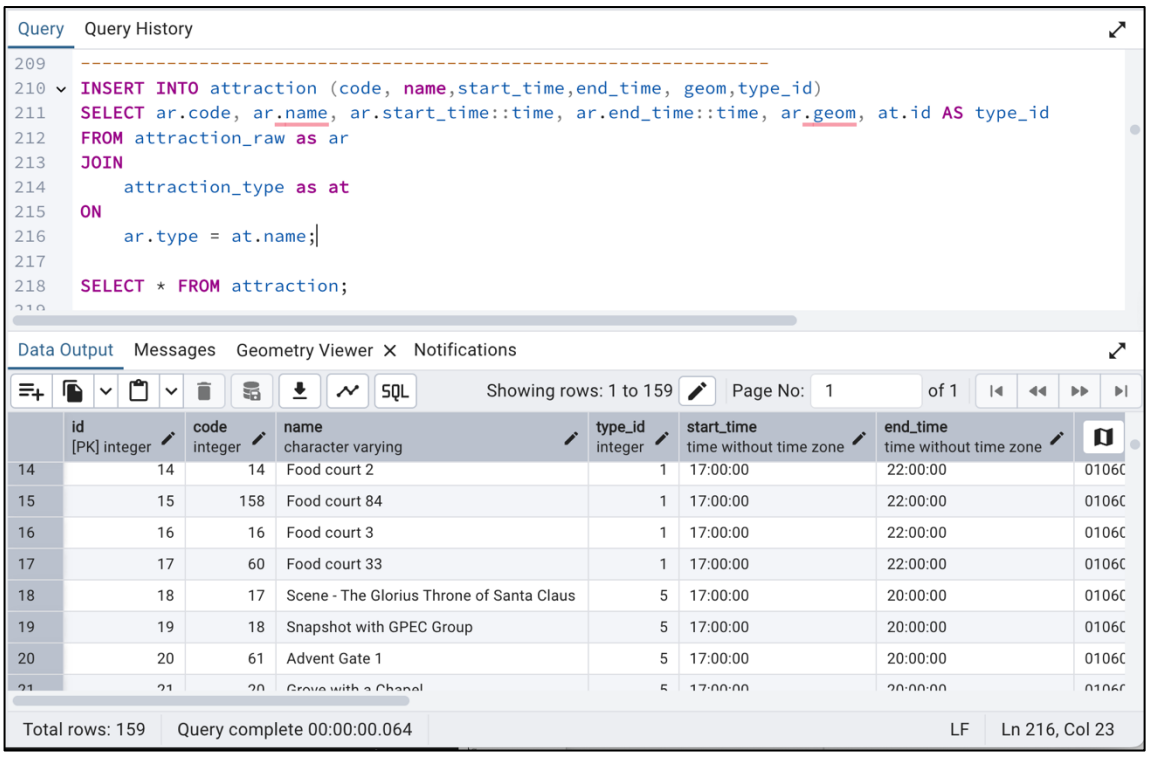
Data Output | Messages | Geometry Viewer | Notifications

Showing rows: 1 to 15 | Page No: 1 of 1

	id [PK] integer	code integer	name character varying	type_id integer	geom geometry
1	1	3	WC2	1	0101000020110F0000470E629DDBAC3F4116753FE0509B5B4
2	2	2	WC1	1	0101000020110F0000C819A376B5AC3F41363B1D1E009B5B4
3	3	5	P3	2	0101000020110F00004C96023BFCAB3F4189A267F1389B5B4
4	4	4	P2	2	0101000020110F00004F4CB78E7CAC3F41EB6F53D0F09A5B4
5	5	1	P1	2	0101000020110F00002BA7F83FE5AC3F41006FFD845B9B5B4
6	6	7	DB4	3	0101000020110F0000BDE71AFBD5AC3F41A9C7F70A4D9B5B.

Total rows: 15 | Query complete 00:00:00.053 | LF | Ln 206, Col 1

Similarly, attraction table is populated using the attraction_raw data and type_id (FK) is extracted from attraction_type.



```
Query  Query History

209  ------------------------------------------------------------
210  INSERT INTO attraction (code, name,start_time,end_time, geom,type_id)
211  SELECT ar.code, ar.name, ar.start_time::time, ar.end_time::time, ar.geom, at.id AS type_id
212  FROM attraction_raw as ar
213  JOIN
214      attraction_type as at
215  ON
216      ar.type = at.name;
217
218  SELECT * FROM attraction;
```

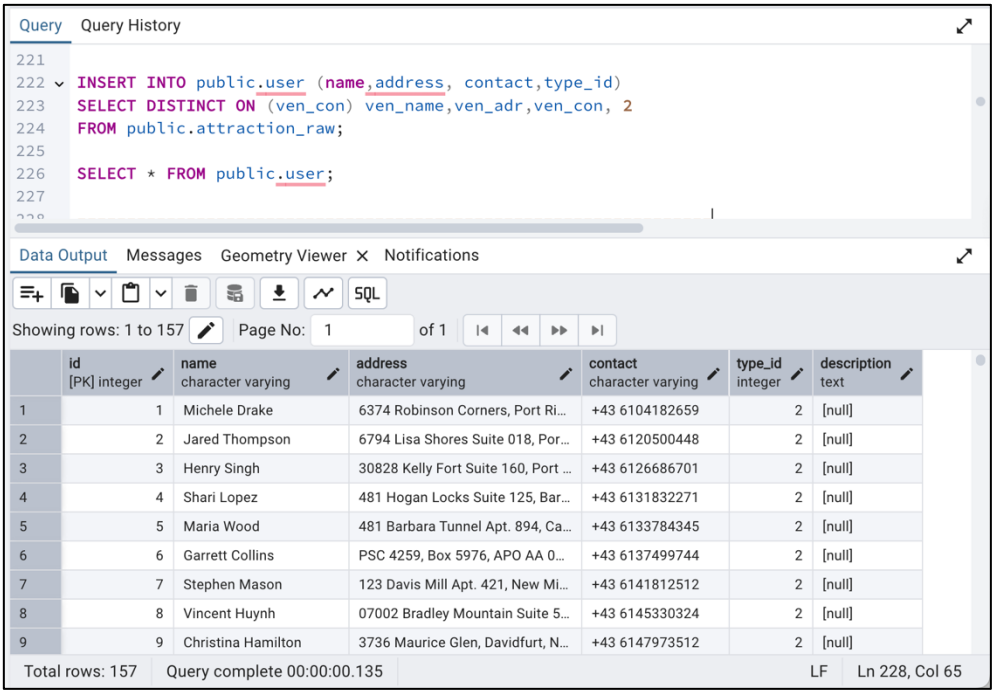
Data Output | Messages | Geometry Viewer | Notifications

Showing rows: 1 to 159 | Page No: 1 of 1

	id [PK] integer	code integer	name character varying	type_id integer	start_time time without time zone	end_time time without time zone	
14	14	14	Food court 2	1	17:00:00	22:00:00	0106(
15	15	158	Food court 84	1	17:00:00	22:00:00	0106(
16	16	16	Food court 3	1	17:00:00	22:00:00	0106(
17	17	60	Food court 33	1	17:00:00	22:00:00	0106(
18	18	17	Scene - The Glorius Throne of Santa Claus	5	17:00:00	20:00:00	0106(
19	19	18	Snapshot with GPEC Group	5	17:00:00	20:00:00	0106(
20	20	61	Advent Gate 1	5	17:00:00	20:00:00	0106(
21	21	20	Grove with a Chapel	5	17:00:00	20:00:00	0106(

Total rows: 159 | Query complete 00:00:00.064 | LF | Ln 216, Col 23

For this case study, contact info is considered a unique identifier of the vendors. Thus, the user table is populated based on unique contacts from the attraction table.



```
Query  Query History

221
222  INSERT INTO public.user (name,address, contact,type_id)
223  SELECT DISTINCT ON (ven_con) ven_name,ven_adr,ven_con, 2
224  FROM public.attraction_raw;
225
226  SELECT * FROM public.user;
227
```

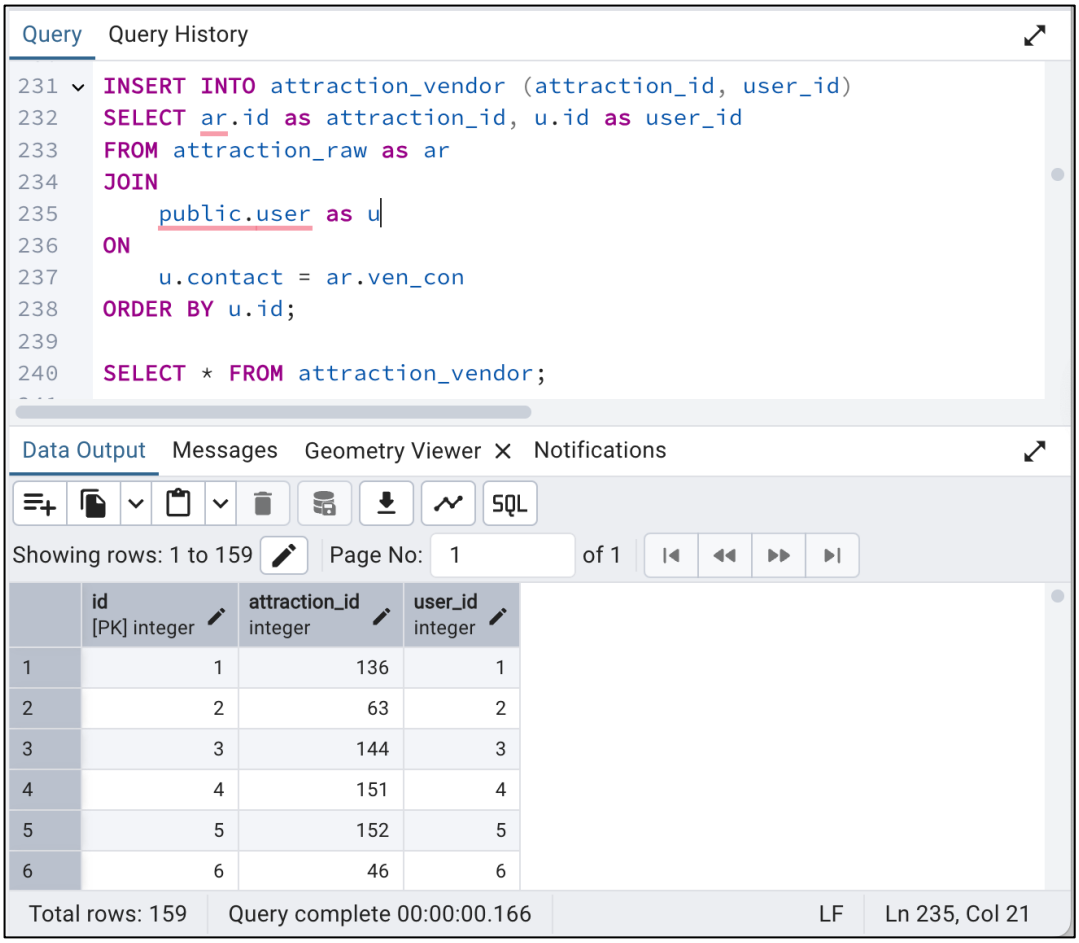
Data Output | Messages | Geometry Viewer | Notifications

Showing rows: 1 to 157 Page No: 1 of 1

	id [PK] integer	name character varying	address character varying	contact character varying	type_id integer	description text
1	1	Michele Drake	6374 Robinson Corners, Port Ri...	+43 6104182659	2	[null]
2	2	Jared Thompson	6794 Lisa Shores Suite 018, Por...	+43 6120500448	2	[null]
3	3	Henry Singh	30828 Kelly Fort Suite 160, Port ...	+43 6126686701	2	[null]
4	4	Shari Lopez	481 Hogan Locks Suite 125, Bar...	+43 6131832271	2	[null]
5	5	Maria Wood	481 Barbara Tunnel Apt. 894, Ca...	+43 6133784345	2	[null]
6	6	Garrett Collins	PSC 4259, Box 5976, APO AA 0...	+43 6137499744	2	[null]
7	7	Stephen Mason	123 Davis Mill Apt. 421, New Mi...	+43 6141812512	2	[null]
8	8	Vincent Huynh	07002 Bradley Mountain Suite 5...	+43 6145330324	2	[null]
9	9	Christina Hamilton	3736 Maurice Glen, Davidfurt, N...	+43 6147973512	2	[null]

Total rows: 157 | Query complete 00:00:00.135 | LF | Ln 228, Col 65

Based on the contacts in attraction_raw and user table, the attraction_vendor table is populated.



```
Query  Query History

231  INSERT INTO attraction_vendor (attraction_id, user_id)
232  SELECT ar.id as attraction_id, u.id as user_id
233  FROM attraction_raw as ar
234  JOIN
235      public.user as u
236  ON
237      u.contact = ar.ven_con
238  ORDER BY u.id;
239
240  SELECT * FROM attraction_vendor;
```

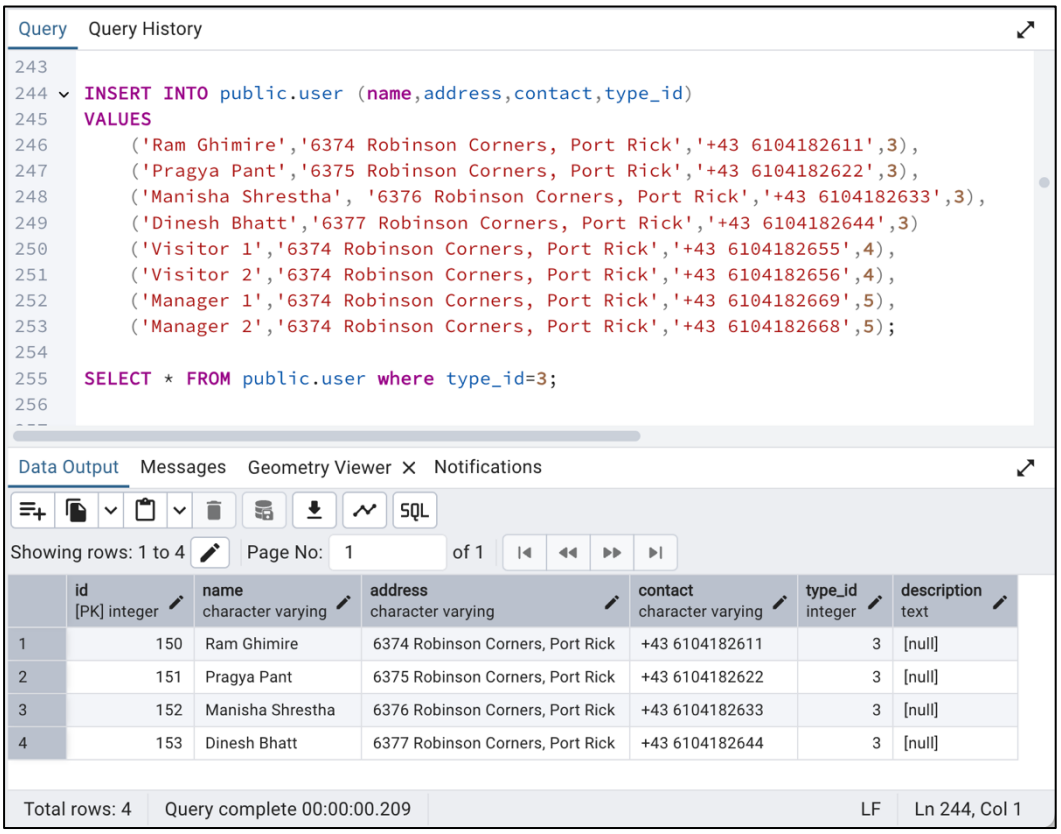
Data Output | Messages | Geometry Viewer | Notifications

Showing rows: 1 to 159 Page No: 1 of 1

	id [PK] integer	attraction_id integer	user_id integer
1	1	136	1
2	2	63	2
3	3	144	3
4	4	151	4
5	5	152	5
6	6	46	6

Total rows: 159 | Query complete 00:00:00.166 | LF | Ln 235, Col 21

The user table consist of only vendors, some other users are added into the table. Here, type_id (FK) determines the type of the user, which is based on user_type table.



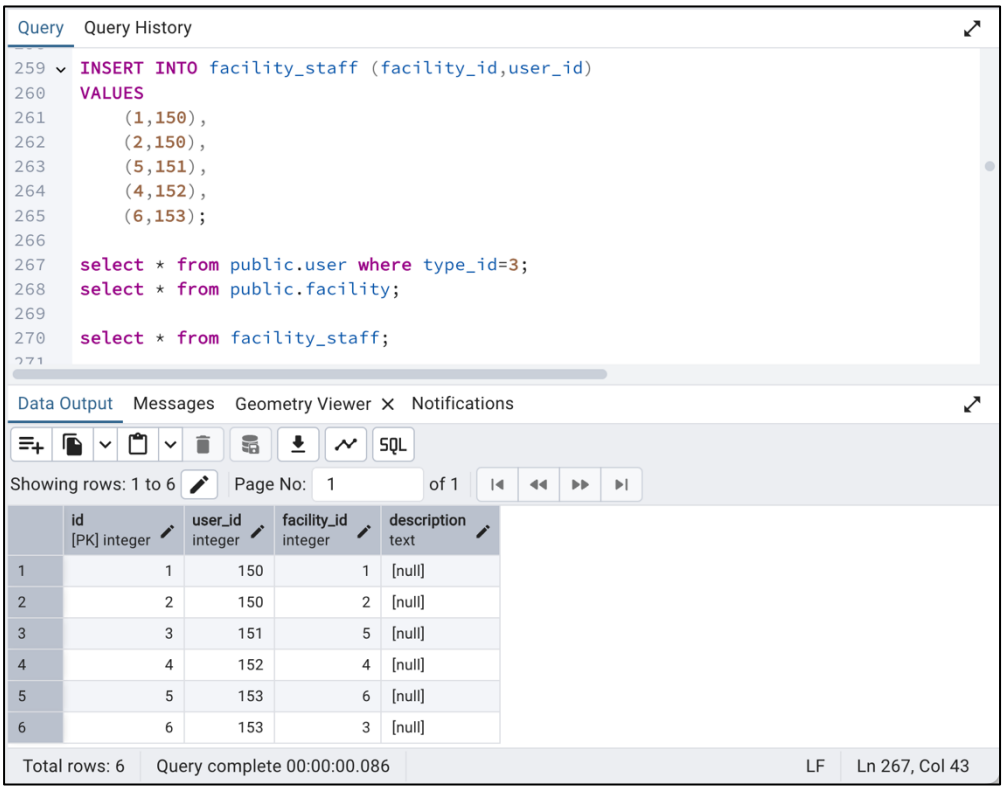
```
INSERT INTO public.user (name,address,contact,type_id)
VALUES
    ('Ram Ghimire','6374 Robinson Corners, Port Rick','+43 6104182611',3),
    ('Pragya Pant','6375 Robinson Corners, Port Rick','+43 6104182622',3),
    ('Manisha Shrestha', '6376 Robinson Corners, Port Rick','+43 6104182633',3),
    ('Dinesh Bhatt','6377 Robinson Corners, Port Rick','+43 6104182644',3)
    ('Visitor 1','6374 Robinson Corners, Port Rick','+43 6104182655',4),
    ('Visitor 2','6374 Robinson Corners, Port Rick','+43 6104182656',4),
    ('Manager 1','6374 Robinson Corners, Port Rick','+43 6104182669',5),
    ('Manager 2','6374 Robinson Corners, Port Rick','+43 6104182668',5);

SELECT * FROM public.user where type_id=3;
```

Showing rows: 1 to 4 Page No: 1 of 1

	id [PK] integer	name character varying	address character varying	contact character varying	type_id integer	description text
1	150	Ram Ghimire	6374 Robinson Corners, Port Rick	+43 6104182611	3	[null]
2	151	Pragya Pant	6375 Robinson Corners, Port Rick	+43 6104182622	3	[null]
3	152	Manisha Shrestha	6376 Robinson Corners, Port Rick	+43 6104182633	3	[null]
4	153	Dinesh Bhatt	6377 Robinson Corners, Port Rick	+43 6104182644	3	[null]

Total rows: 4 Query complete 00:00:00.209 LF Ln 244, Col 1

Based on user table and facility table, following values are inserted into facility_staff table that define the role for particular staff.



```
INSERT INTO facility_staff (facility_id,user_id)
VALUES
    (1,150),
    (2,150),
    (5,151),
    (4,152),
    (6,153);

select * from public.user where type_id=3;
select * from public.facility;

select * from facility_staff;
```

Showing rows: 1 to 6 Page No: 1 of 1

	id [PK] integer	user_id integer	facility_id integer	description text
1	1	150	1	[null]
2	2	150	2	[null]
3	3	151	5	[null]
4	4	152	4	[null]
5	5	153	6	[null]
6	6	153	3	[null]

Total rows: 6 Query complete 00:00:00.086 LF Ln 267, Col 43

4.7 Create View

The following views were created. Attraction_view and facility_view are again denormalized by defining the column type based on the type_if (FK) field.

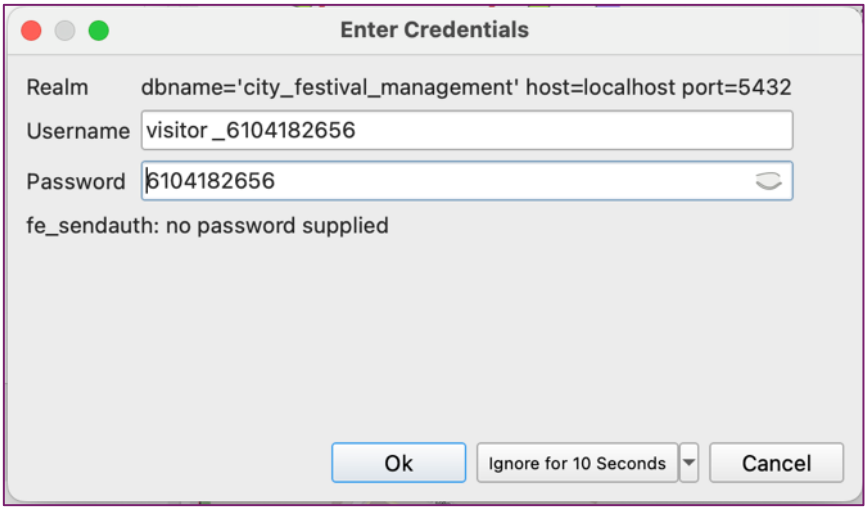
```
CREATE VIEW attraction_view AS
    SELECT

a.id,a.name,a.start_date,a.end_date,a.start_time,a.end_time,a.geom,a.d
escription,at.name as type
    FROM
        attraction as a
    JOIN
        attraction_type as at
    ON
        a.type_id=at.id;

CREATE VIEW facility_view AS
    SELECT
        f.id,f.name,f.geom,ft.name as type
    FROM
        facility as f
    JOIN
        facility_type as ft
    ON
        f.type_id=ft.id;

CREATE VIEW pathway_view AS
    SELECT
        *
    FROM
        pathway;
```

4.8 Define Roles and Permissions

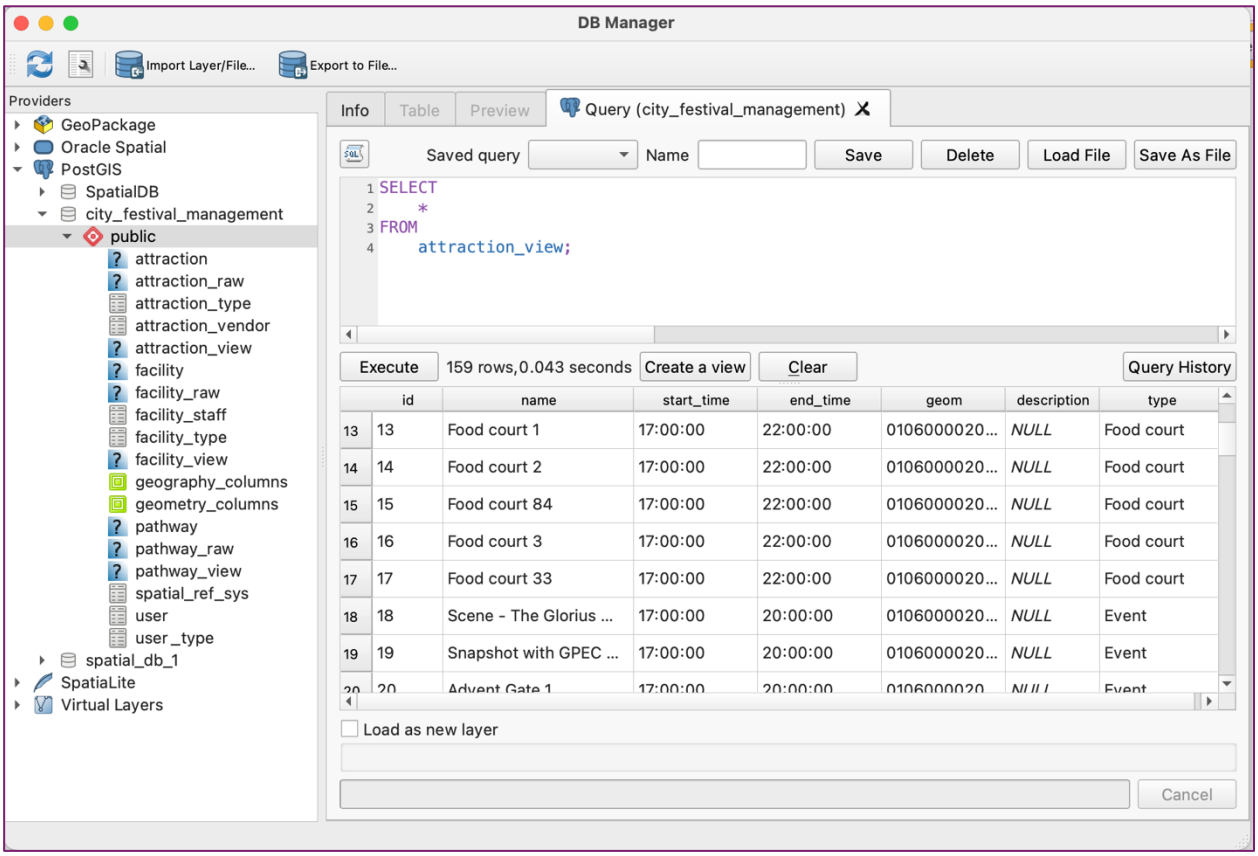
The visitor role was created for one of the visitor users using a contact number, which is a unique column. Only view access is provided for the attraction_view, facility_view, pathway_view, attraction_type, and facility_type.

```
CREATE ROLE visitor_6104182656 LOGIN PASSWORD '6104182656';

DO $$
DECLARE
    role_name TEXT := 'visitor_6104182656';
BEGIN
    EXECUTE format('REVOKE ALL ON attraction_view FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON attraction_view TO %I', role_name);

    EXECUTE format('REVOKE ALL ON facility_view FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON facility_view TO %I', role_name);

    EXECUTE format('REVOKE ALL ON pathway_view FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON pathway_view TO %I', role_name);

    EXECUTE format('REVOKE ALL ON attraction_type FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON attraction_type TO %I', role_name);

    EXECUTE format('REVOKE ALL ON facility_type FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON facility_type TO %I', role_name);
END $$;
```

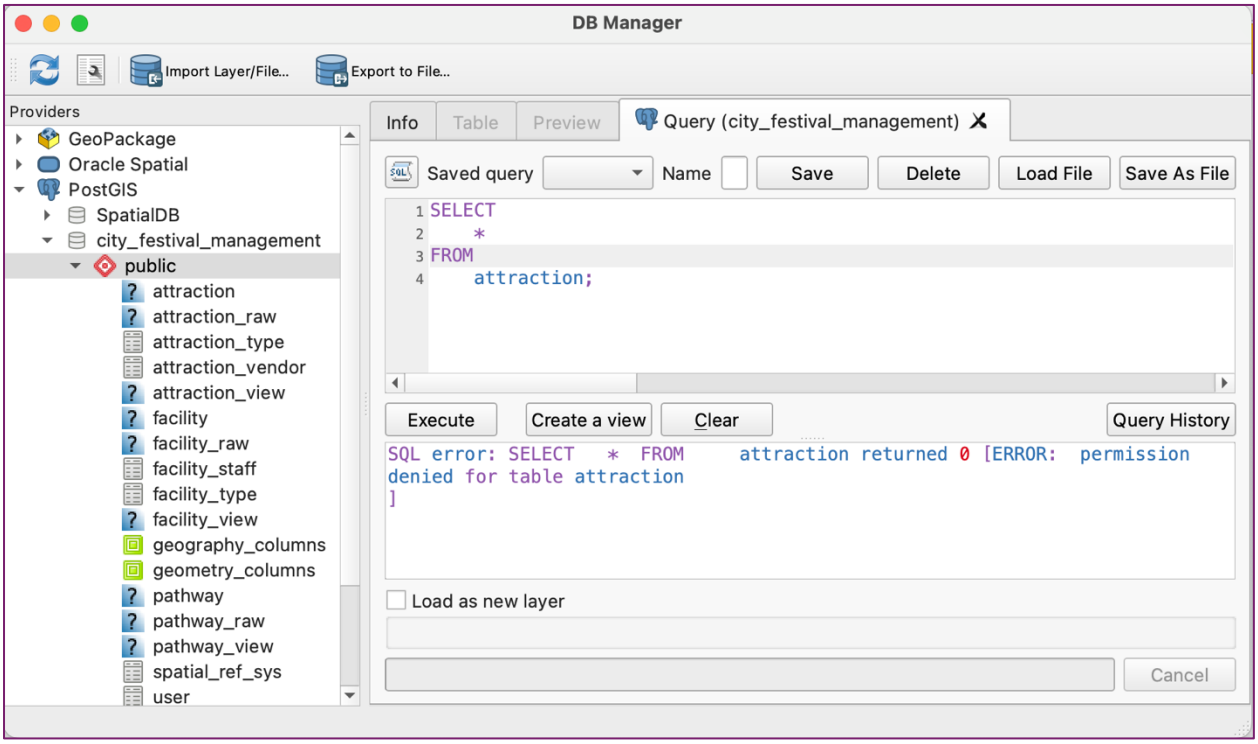
Then, the database is connected with the visitor credentials as shown below.



As a visitor, the attraction view is accessible.



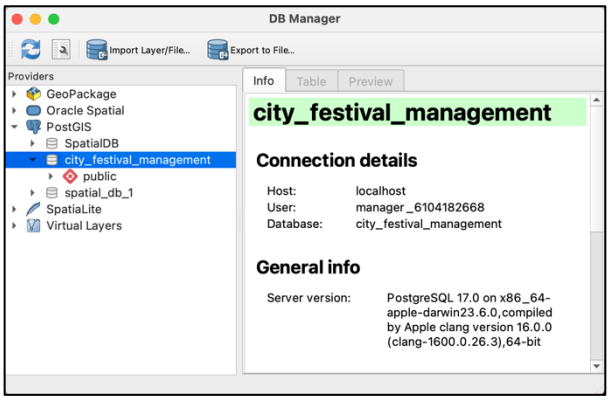
But, the attraction table is not accessible as visitors have no permission to access this table.



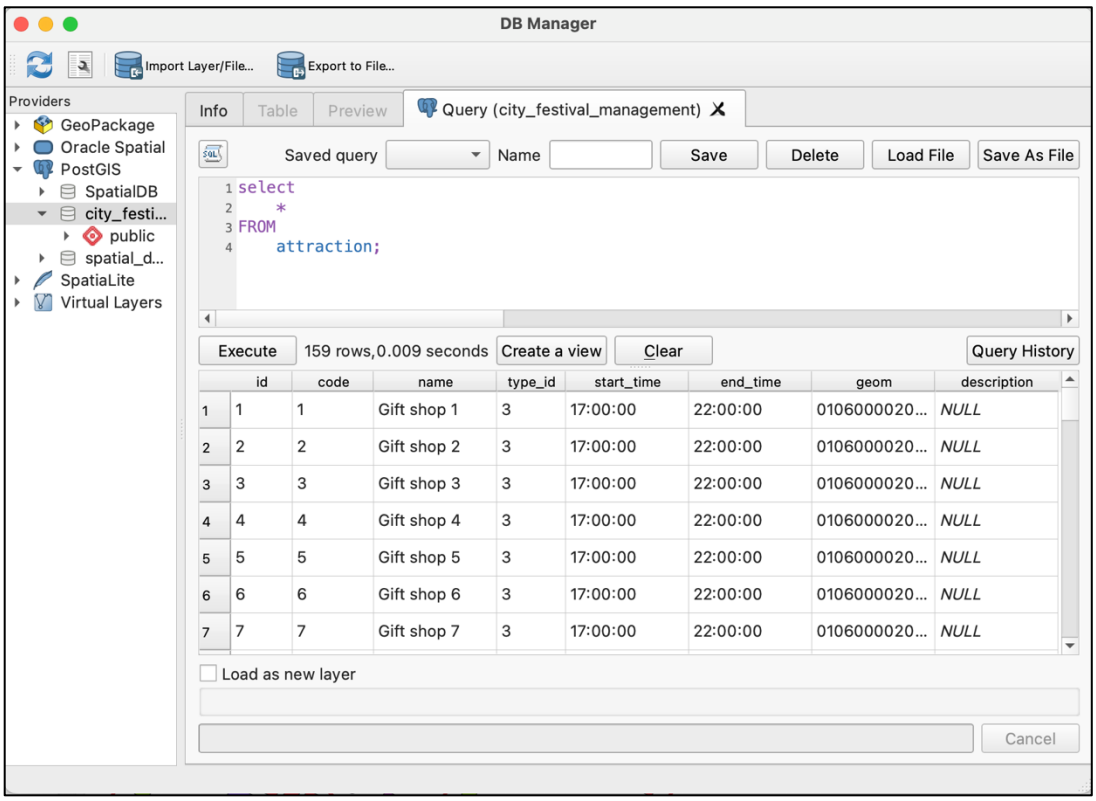
Likewise, the **manager role** was created for the manager user added in the user table with permission, as shown below.

```
CREATE ROLE manager_6104182668 LOGIN PASSWORD '6104182668';
DO $$
DECLARE
    role_name TEXT := 'manager_6104182668';
BEGIN
    EXECUTE format('REVOKE ALL ON attraction FROM %I', role_name);
    EXECUTE format('GRANT USAGE, SELECT ON SEQUENCE attraction_id_seq TO %I', role_name);
    EXECUTE format('GRANT SELECT, INSERT, UPDATE, DELETE ON attraction TO %I', role_name);

    EXECUTE format('REVOKE ALL ON facility FROM %I', role_name);
    EXECUTE format('GRANT USAGE, SELECT ON SEQUENCE facility_id_seq TO %I', role_name);
    EXECUTE format('GRANT SELECT, INSERT, UPDATE, DELETE ON facility TO %I', role_name);

    EXECUTE format('REVOKE ALL ON pathway FROM %I', role_name);
    EXECUTE format('GRANT USAGE, SELECT ON SEQUENCE pathway_id_seq TO %I', role_name);
    EXECUTE format('GRANT SELECT, INSERT, UPDATE, DELETE ON pathway TO %I', role_name);

    EXECUTE format('REVOKE ALL ON attraction_type FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON attraction_type TO %I', role_name);

    EXECUTE format('REVOKE ALL ON facility_type FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON facility_type TO %I', role_name);

    EXECUTE format('REVOKE ALL ON public.user FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON public.user TO %I', role_name);

    EXECUTE format('REVOKE ALL ON attraction_vendor FROM %I', role_name);
    EXECUTE format('GRANT USAGE, SELECT ON SEQUENCE attraction_vendor_id_seq TO %I', role_name);
    EXECUTE format('GRANT SELECT, INSERT, UPDATE, DELETE ON attraction_vendor TO %I', role_name);

    EXECUTE format('REVOKE ALL ON facility_staff FROM %I', role_name);
    EXECUTE format('GRANT USAGE, SELECT ON SEQUENCE facility_staff_id_seq TO %I', role_name);
    EXECUTE format('GRANT SELECT, INSERT, UPDATE, DELETE ON facility_staff TO %I', role_name);

    EXECUTE format('REVOKE ALL ON attraction_view FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON attraction_view TO %I', role_name);

    EXECUTE format('REVOKE ALL ON facility_view FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON facility_view TO %I', role_name);

    EXECUTE format('REVOKE ALL ON pathway_view FROM %I', role_name);
    EXECUTE format('GRANT SELECT ON pathway_view TO %I', role_name);

END $$;
```

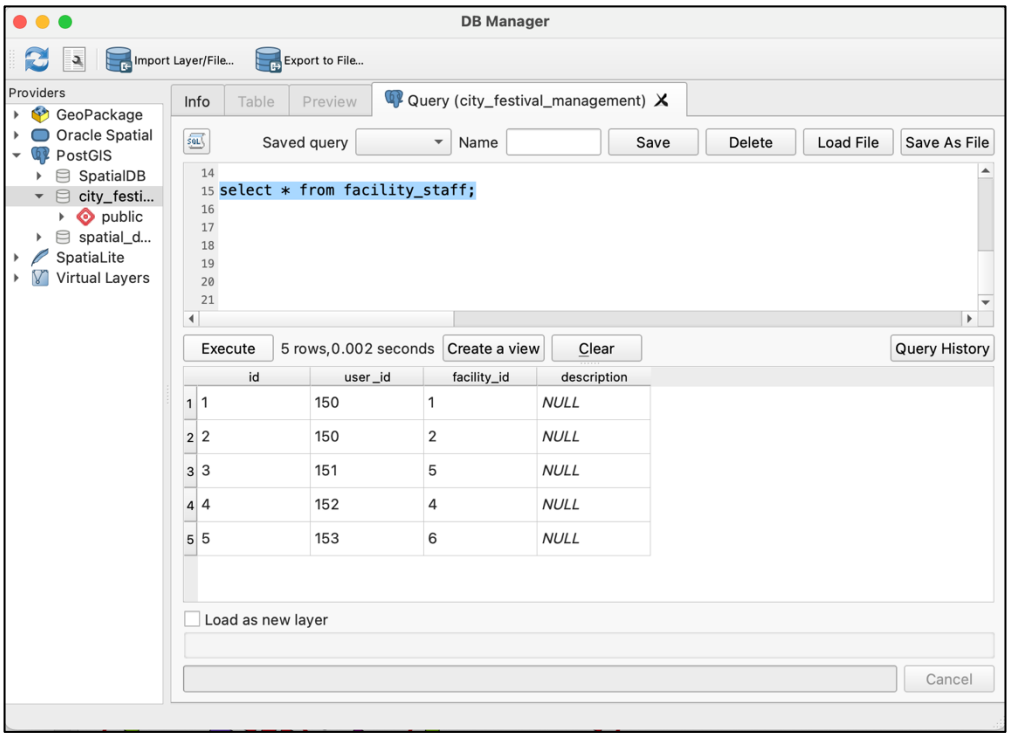
Then, database is connected with the visitor credential as shown below.

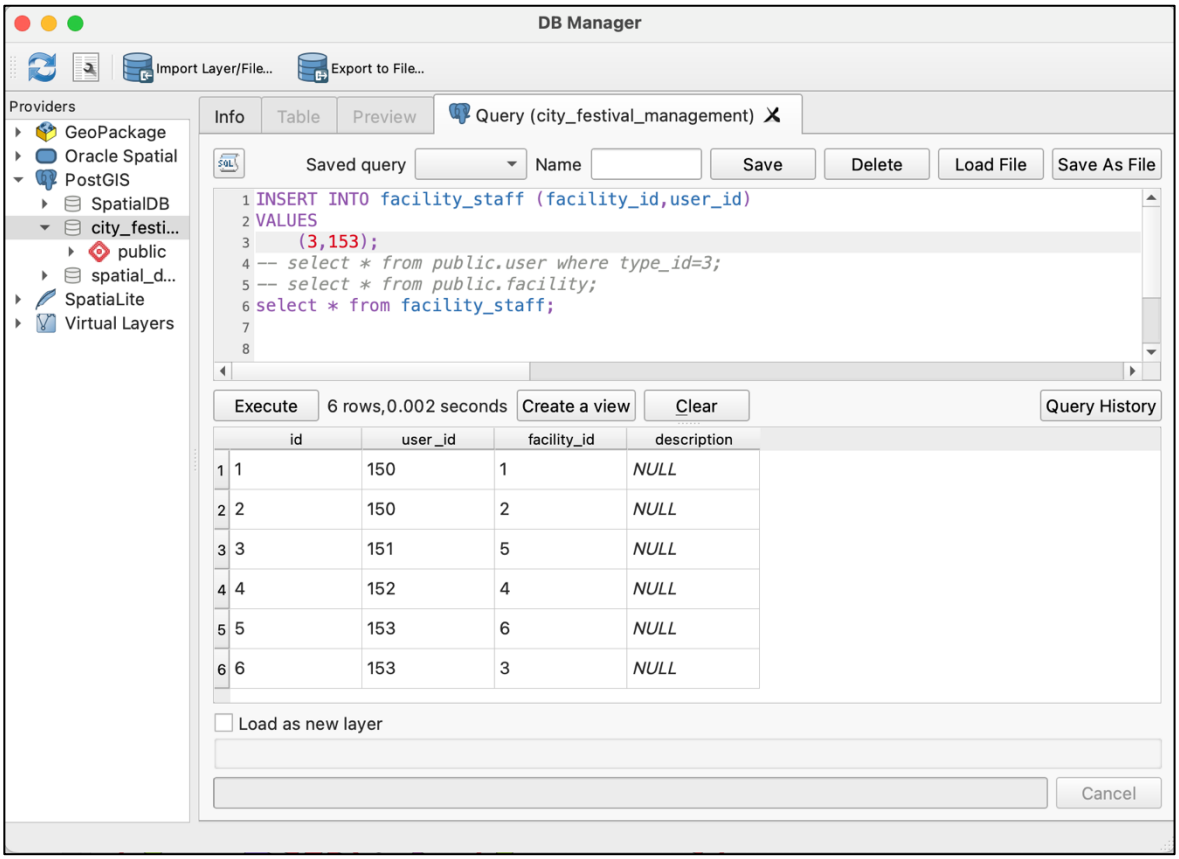


As a manager, the user can access the tables and views as well.



Manager role users also can add, update, and delete the records. For instance, the manager can add the information regarding the staff and facility responsibility, as shown below.





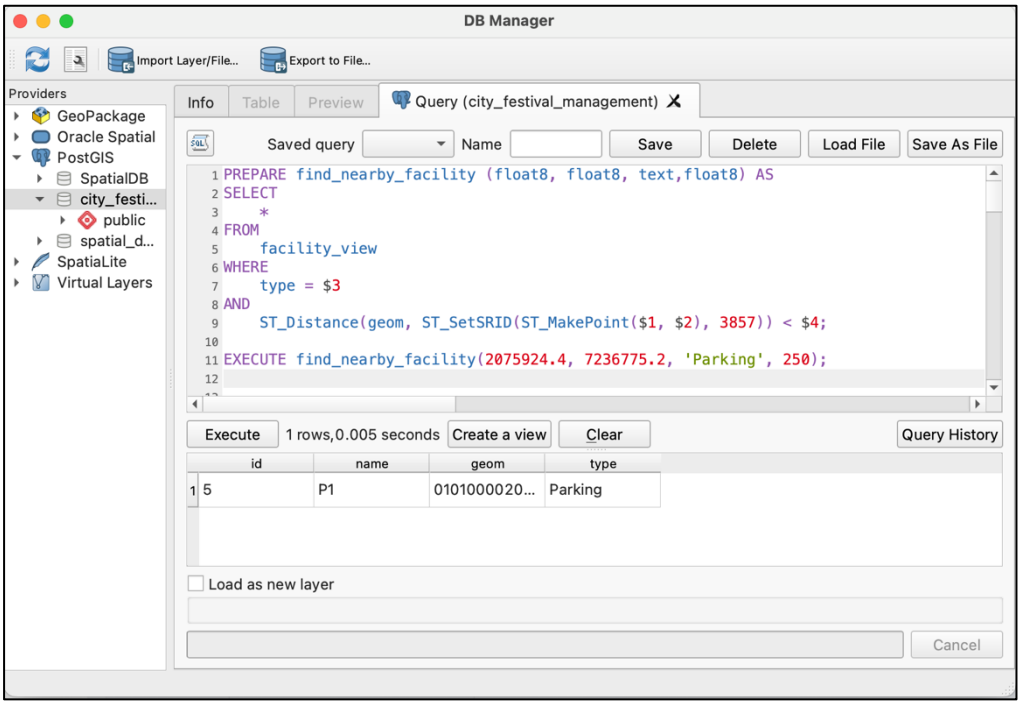
In this way, users with manager roles can manipulate the data with the database depending on the access given to the particular tables.

5 QUERIES

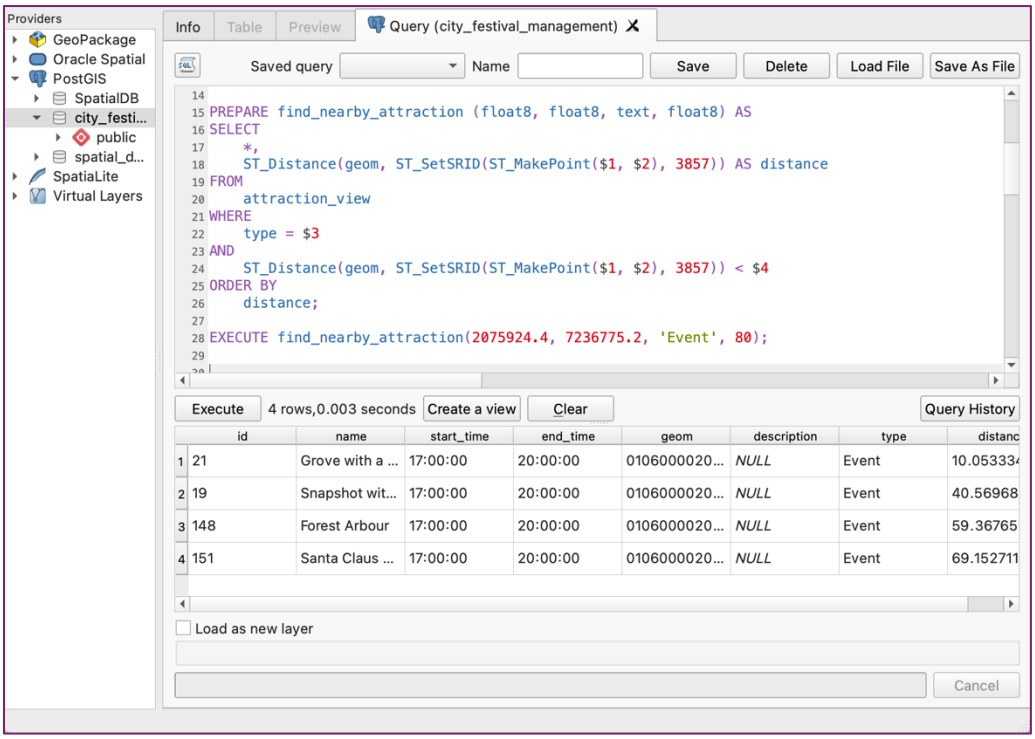
5.1 Queries based on User's Location

a. To find the list of nearby facilities based on user's location, facility_type, and distance parameter.

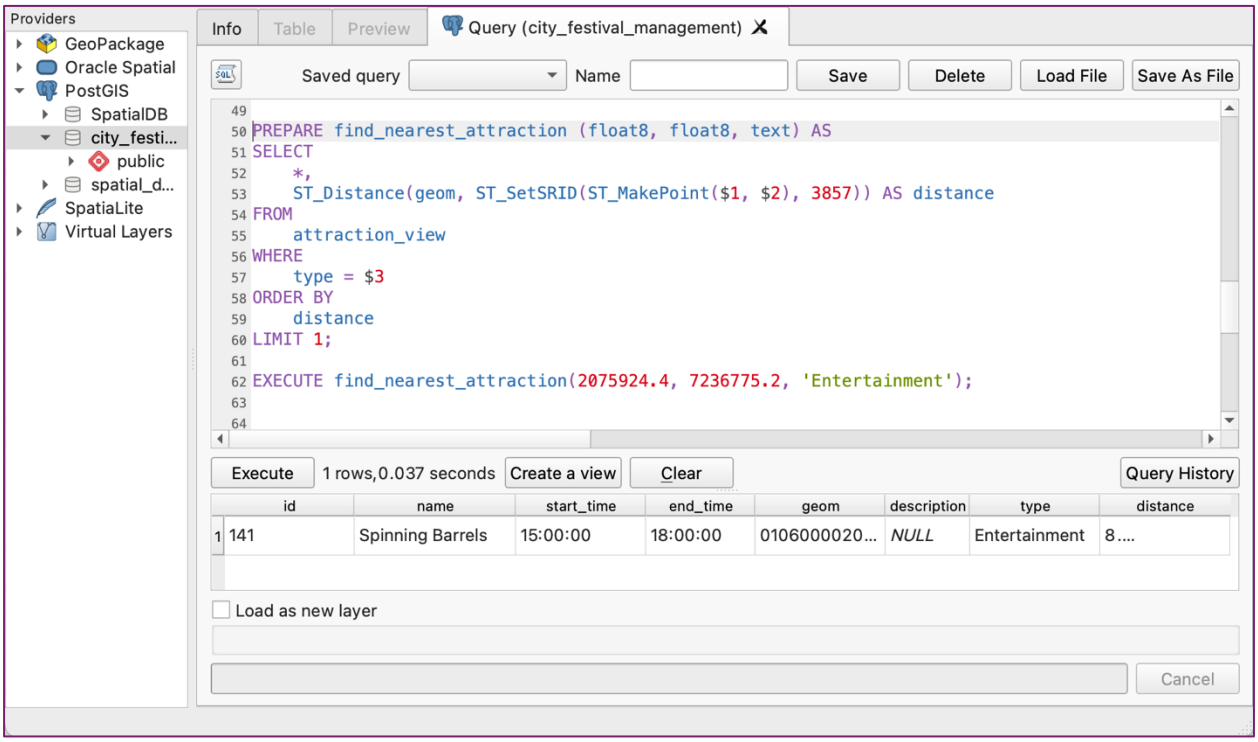
PREPARE statement was defined that takes location (lat & lon), facility_type, and within_distance value as parameters and return the list of nearby facilities.



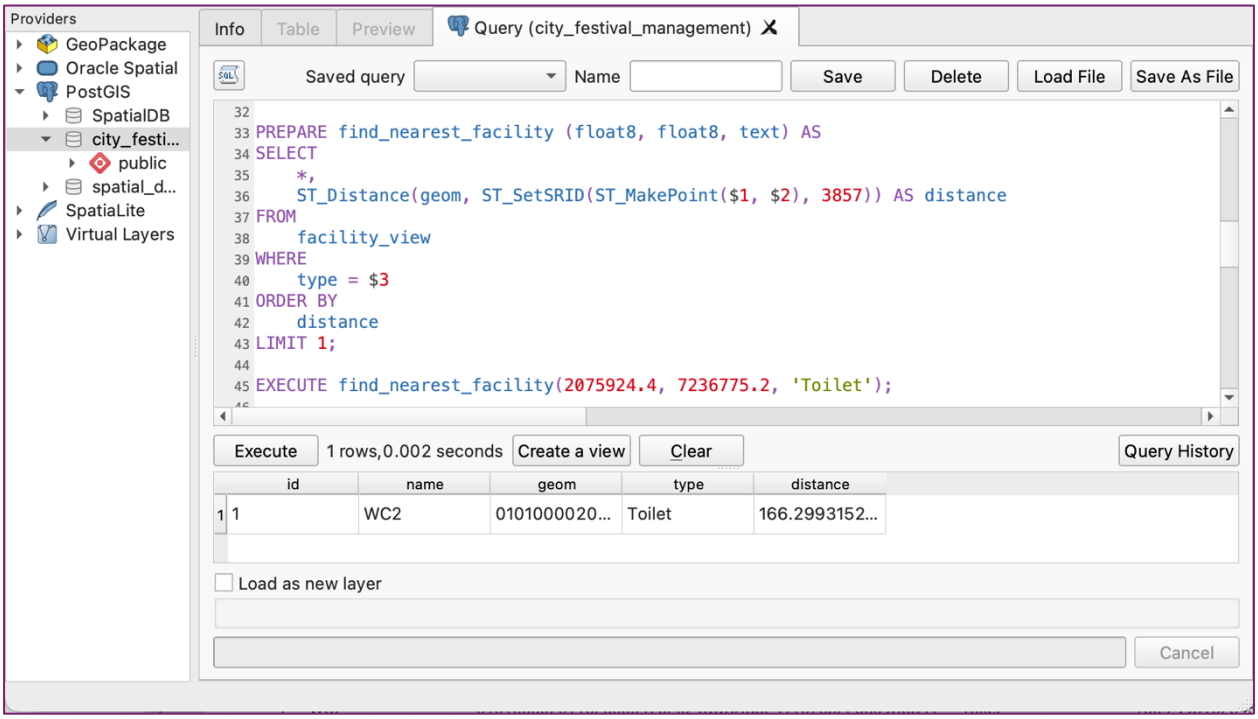
b. To find the list of nearby attractions based on user's location, attraction_type, and distance parameter.



c. To find the nearest attraction based on the user's location and attraction_type parameter

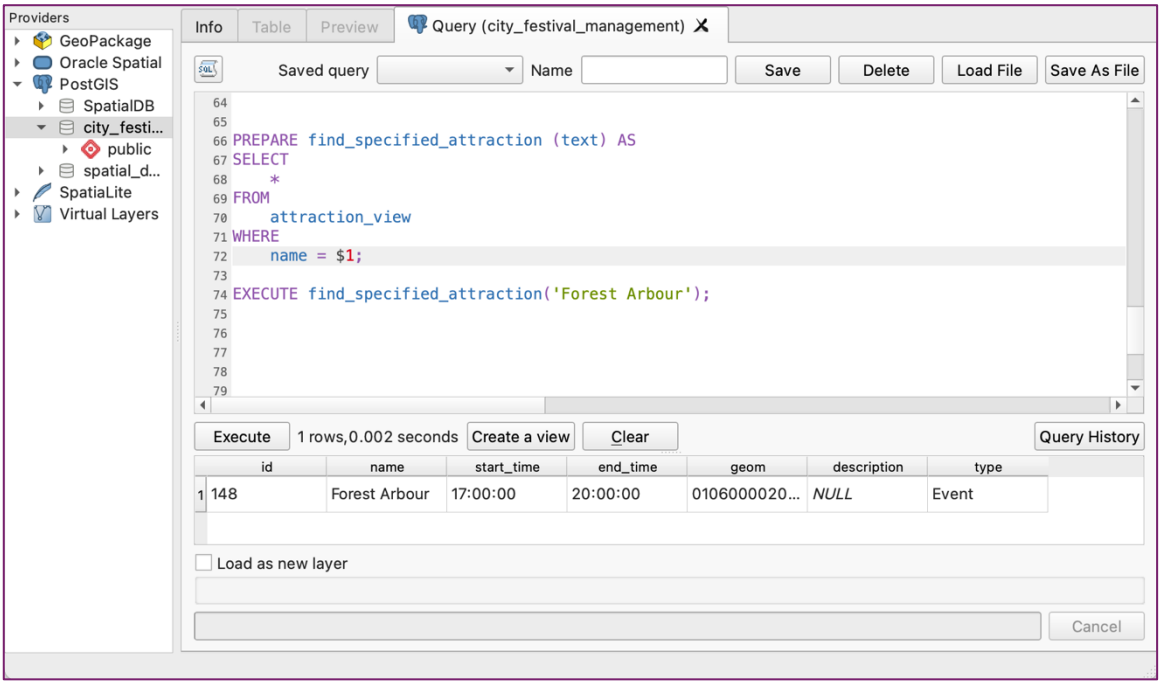


d. To find the nearest facility based on user's location and facility_type parameter

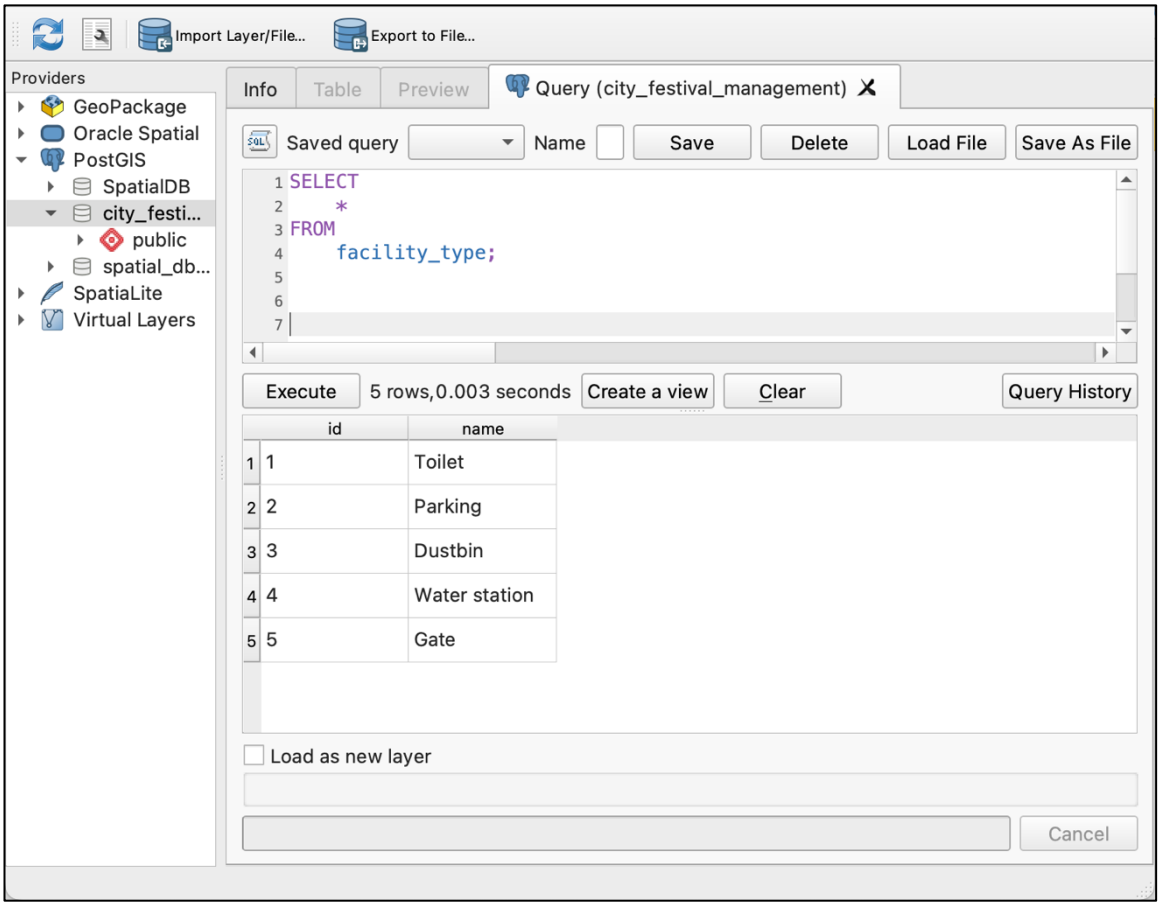


5.2 Queries not based on User's Location

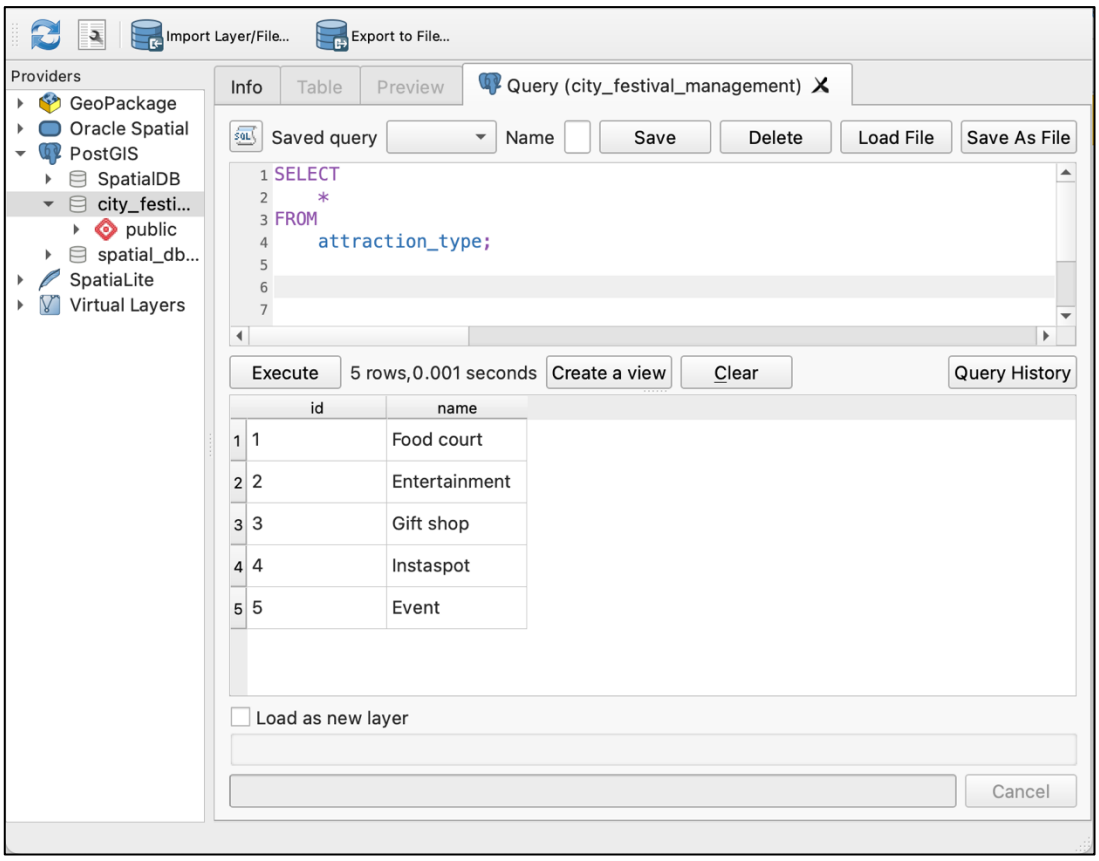
a. To get the specific attraction or facility



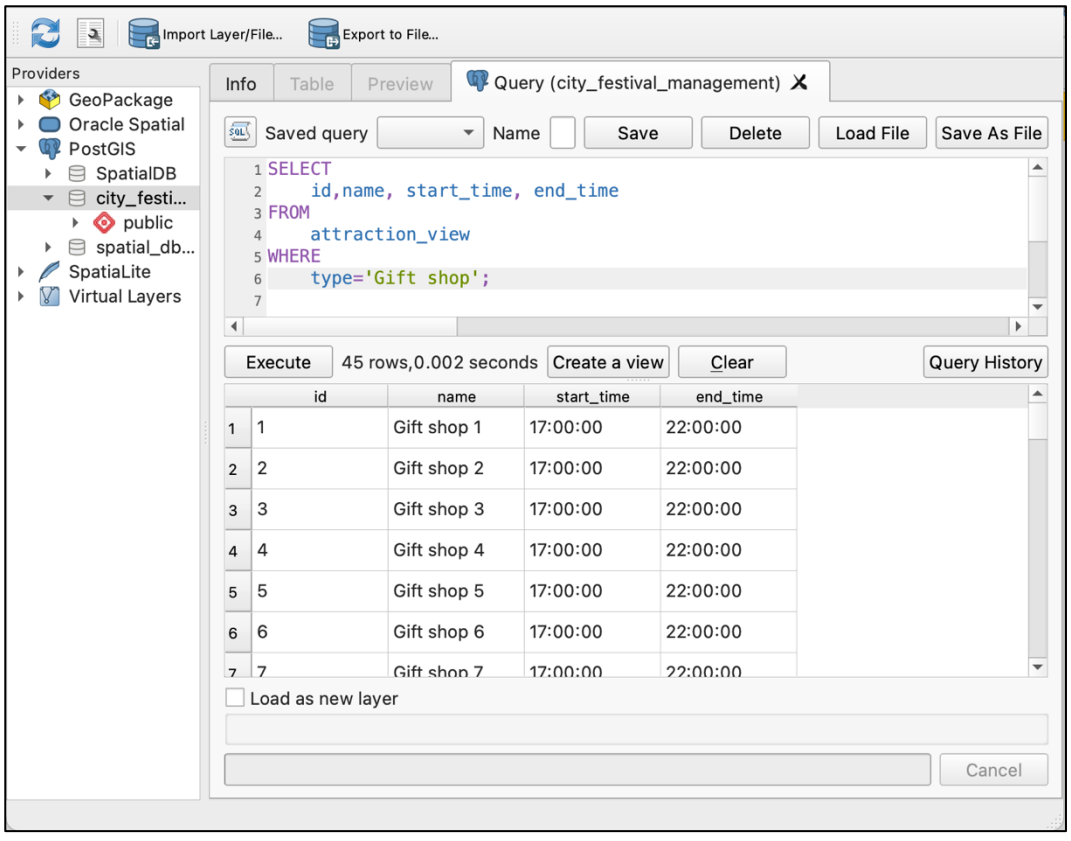
b. To get list of facility types



c. To get list of attraction types



d. To get all the lists of attractions or facilities based on type



6 PGROUTING IMPLEMENTATION

6.1 Installation and create extension

Following the steps mentioned in this source, pgrouting extension was installed as follows:

```
>> git clone https://github.com/pgRouting/pgrouting.git
>> cd pgrouting
>> mkdir build
>> cd build
>> cmake ..
>> make
>> sudo make install
>> /Library/PostgreSQL/17/bin/pg_ctl -D /Library/PostgreSQL/17/data restart
```

```
CREATE EXTENSION pgrouting;
```

6.2 Building Routing Topology

Multilinestring geometry type was not suppported in pgrouting, thus pathway_edges table of linetring geomtery data type was created from pathway table as folows:

```
SELECT DISTINCT ST_GeometryType(geom)
FROM pathway;

CREATE TABLE pathway_edges (
    id SERIAL PRIMARY KEY,
    name TEXT,
    geom GEOMETRY(LINESTRING, 3857) NOT NULL
);


INSERT INTO pathway_edges (name, geom)
SELECT  name, (ST_Dump(geom)).geom AS geom
FROM pathway;
```

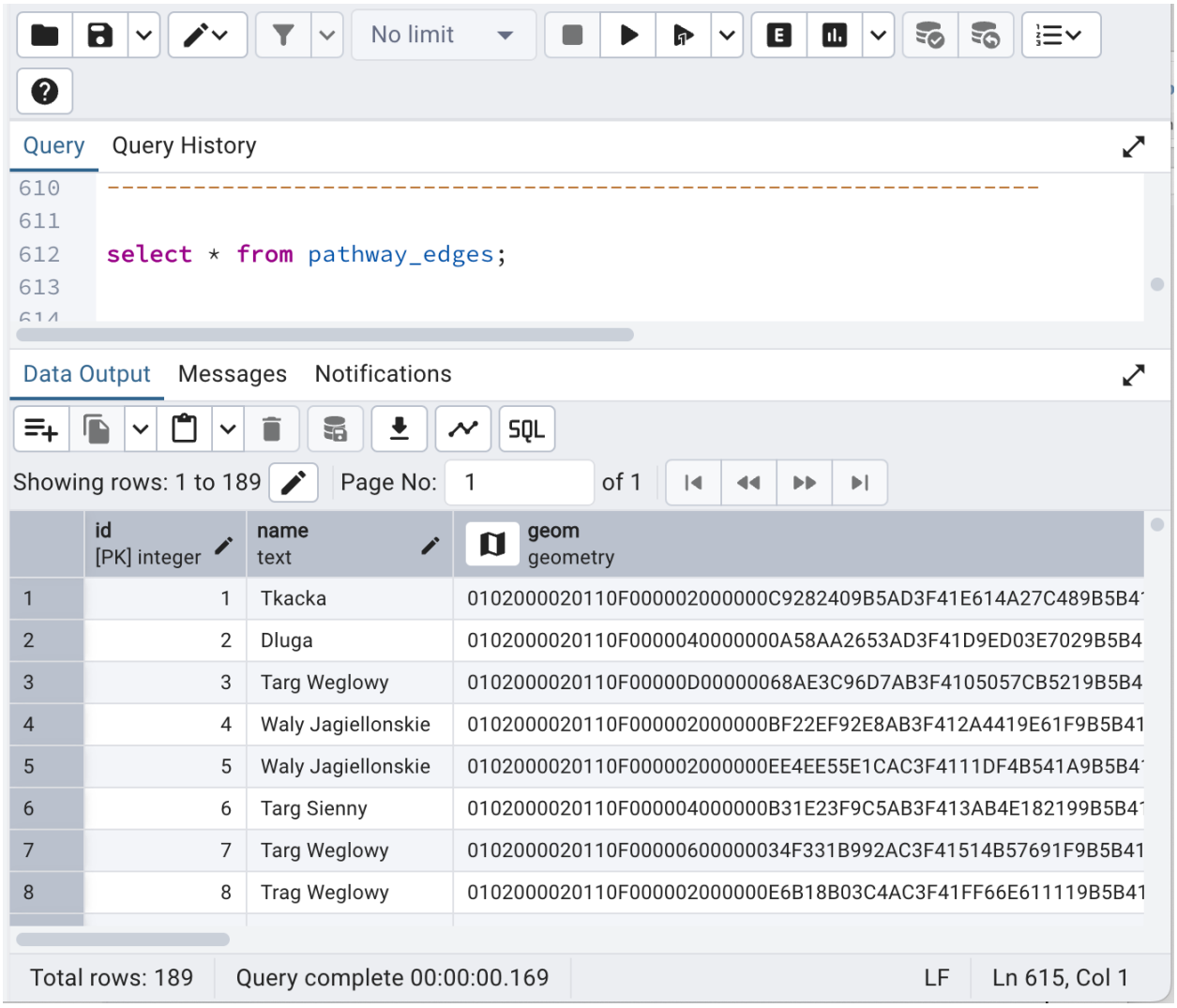
Since all the facility points are not linked with the pathway, following SQL was executed to create a new line that connects the facility points to the nearest pathway.

```
INSERT INTO pathway_edges (geom)
SELECT
    ST_MakeLine(facility.geom,
                (SELECT ST_ClosestPoint(roads.geom, facility.geom)
                 FROM pathway_edges AS roads
                 ORDER BY ST_Distance(roads.geom, facility.geom)
                 LIMIT 1)
               ) AS geom

FROM facility;
```

Similarly, to connect attractions to the nearest pathway, centroid of the polygon is created and connects between centroid and nearest pathway as follow:

```
ALTER TABLE attraction
ADD COLUMN centroid geometry(Point, 3857);

UPDATE attraction
SET centroid = ST_Centroid(geom);
------------------
INSERT INTO pathway_edges (geom)
SELECT
    ST_MakeLine(attraction.centroid,
                (SELECT ST_ClosestPoint(roads.geom, attraction.centroid)
                 FROM pathway_edges AS roads
                 ORDER BY ST_Distance(roads.geom, attraction.centroid)
                 LIMIT 1)
               ) AS geom
FROM attraction;
```

In total, 189 edges is formed that contain original pathways, connections between facilities and pathways, and attractions and pathways.

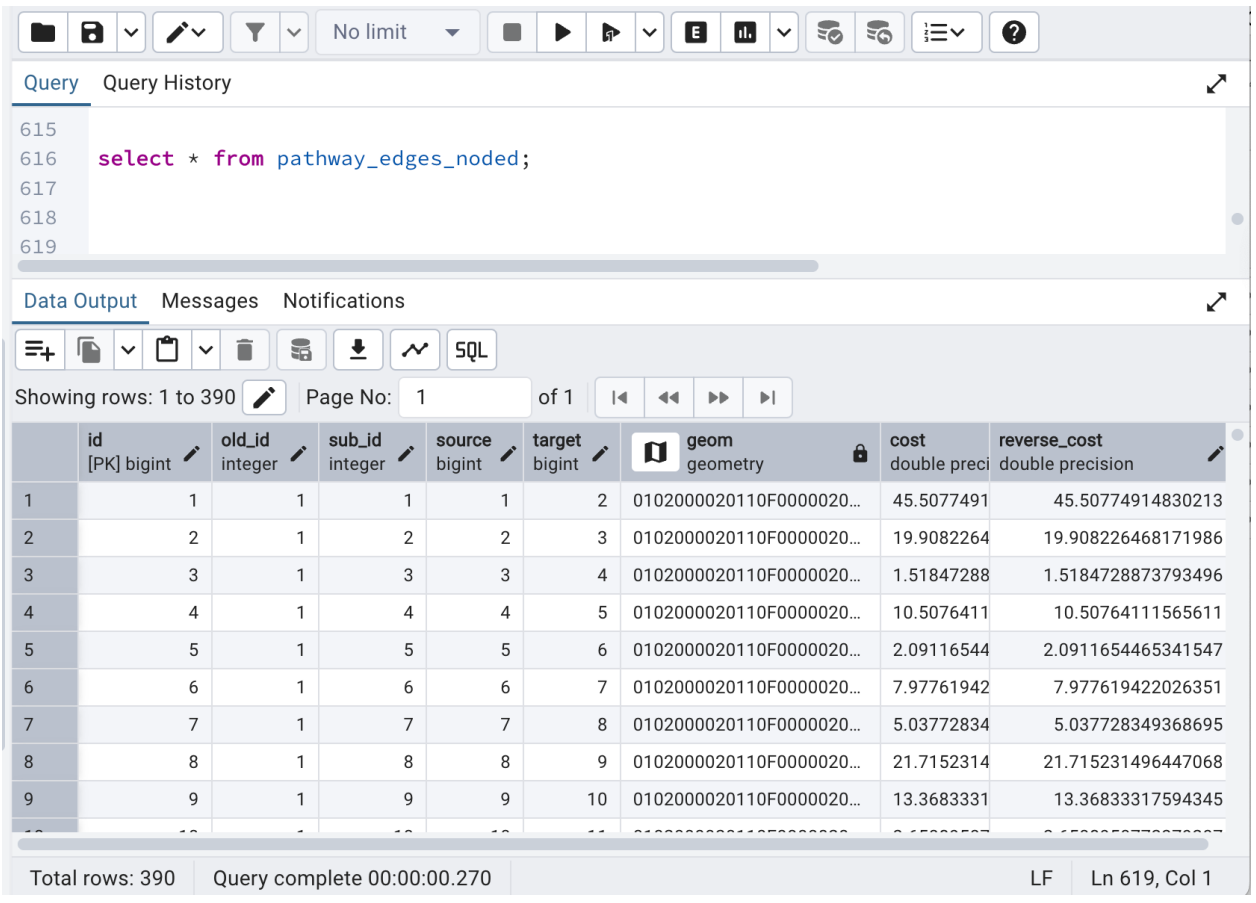


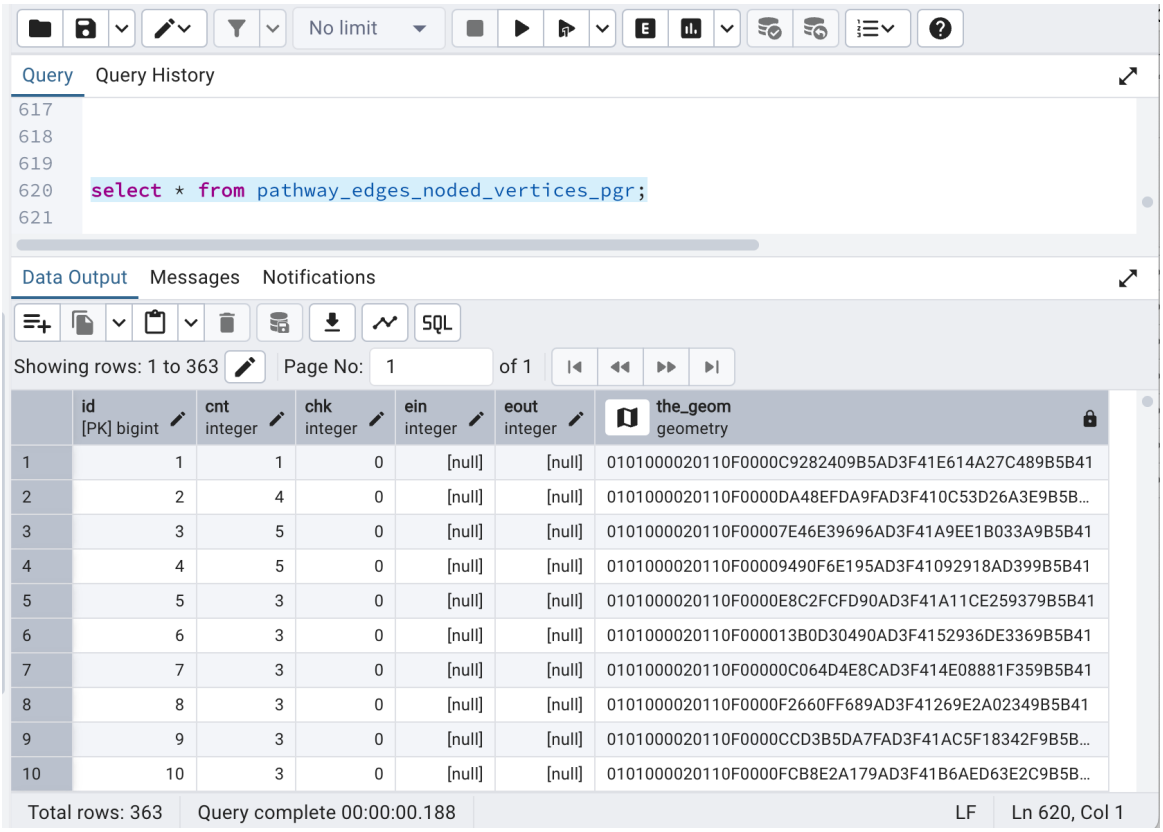
The **pathway_edges_noded** table is created after executing the following SQL query. This command basically processes an existing **edges table** (e.g., pathway_edges) to introduce **nodes** at intersections or breakpoints. This transformation ensures that the pathway network properly represents connectivity at junctions.

```
SELECT pgr_nodeNetwork('pathway_edges', 0.5, the_geom => 'geom');
```

Four new columns are added to the pathway_edges_noded table to store essential information for network analysis and routing. These columns typically include the source (starting node) and target (ending node) of each edge. After adding these columns, the pgr_createTopology SQL function is executed to populate the source and target columns based on the geometry of the pathways. It also creates a new table "pathway_edges_noded_vertices_pgr", which represents the network topology in a vertex-edge format. This table is important to calculate the shortest path or optimal routes in the network.

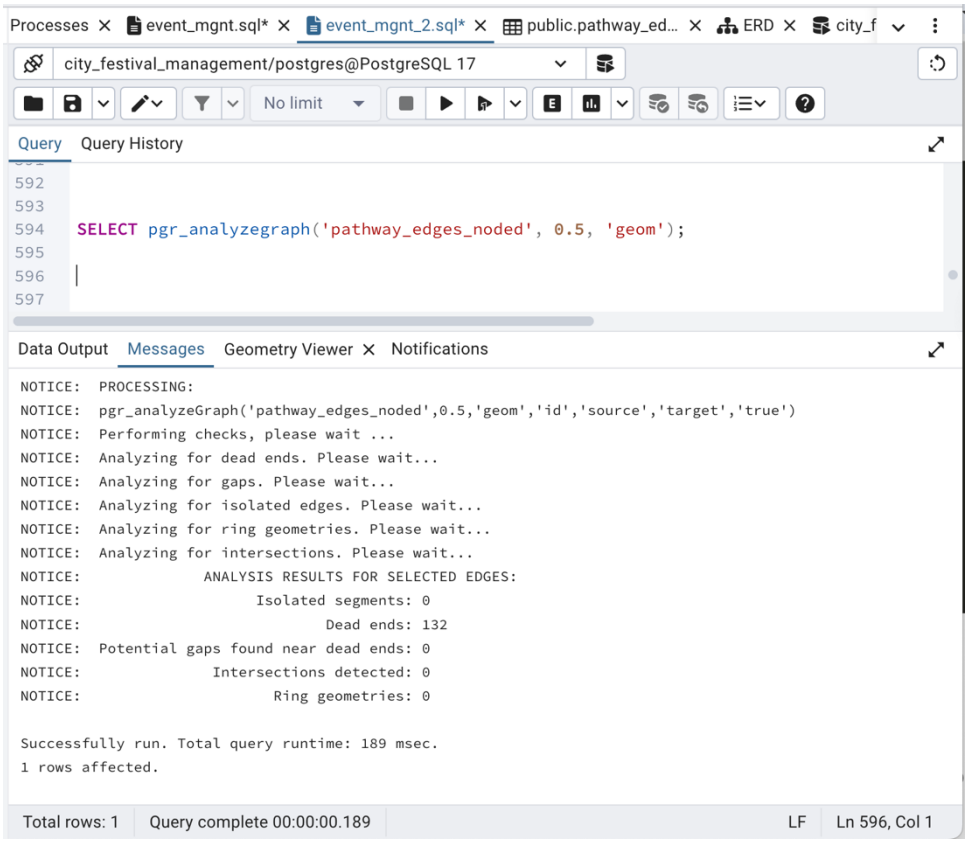
```
ALTER TABLE pathway_edges_noded ADD COLUMN source INTEGER;
ALTER TABLE pathway_edges_noded ADD COLUMN target INTEGER;
ALTER TABLE pathway_edges_noded ADD COLUMN cost DOUBLE PRECISION;
ALTER TABLE pathway_edges_noded ADD COLUMN reverse_cost DOUBLE PRECISION;
--------------------------------------------------------------------------------

SELECT pgr_createTopology('pathway_edges_noded', 0.5, 'geom', 'id');

--------------------------------------------------------------------------------

UPDATE pathway_edges_noded SET
cost = ST_length(geom),
reverse_cost =ST_length(geom);
```





	id [PK] bigint	cnt integer	chk integer	ein integer	eout integer	the_geom geometry
1	1	1	0	[null]	[null]	0101000020110F0000C9282409B5AD3F41E614A27C489B5B41
2	2	4	0	[null]	[null]	0101000020110F0000DA48EFDA9FAD3F410C53D26A3E9B5B...
3	3	5	0	[null]	[null]	0101000020110F00007E46E39696AD3F41A9EE1B033A9B5B41
4	4	5	0	[null]	[null]	0101000020110F00009490F6E195AD3F41092918AD399B5B41
5	5	3	0	[null]	[null]	0101000020110F0000E8C2FCFD90AD3F41A11CE259379B5B41
6	6	3	0	[null]	[null]	0101000020110F000013B0D30490AD3F4152936DE3369B5B41
7	7	3	0	[null]	[null]	0101000020110F00000C064D4E8CAD3F414E08881F359B5B41
8	8	3	0	[null]	[null]	0101000020110F0000F2660FF689AD3F41269E2A02349B5B41
9	9	3	0	[null]	[null]	0101000020110F0000CCD3B5DA7FAD3F41AC5F18342F9B5B...
10	10	3	0	[null]	[null]	0101000020110F0000FCB8E2A179AD3F41B6AED63E2C9B5B...

pgr_analyzeGraph is executed to analyze the network topology. It returns OK as an output if the analysis has finished with the information as shown in the figure below, and it returns FAIL when the analysis was not completed due to an error.



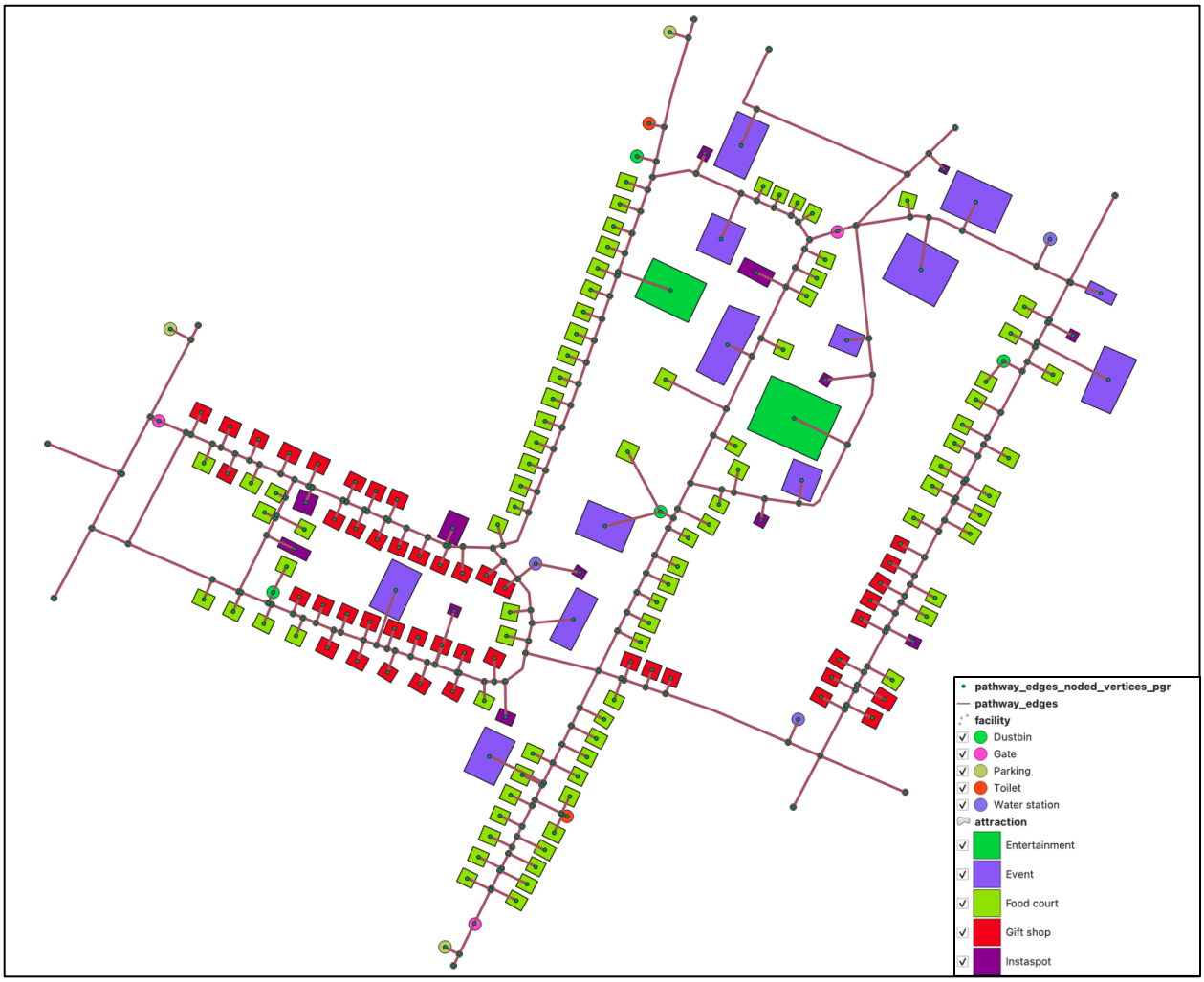


Figure 12:Network Topology Map

6.3 Routing Queries

a. Route nearest facility based on facility_type

Pgr_dijkstra algorithm was used to calculate the shortest path. It is a graph search algorithm that solves the shortest path problem for a graph with non-negative edge path costs, producing a shortest path from a starting vertex to an ending vertex. This implementation can be used with a directed graph and an undirected graph. The following PREPARE statement is defined to calculate and generate the shortest path to the nearest facility. The facility_type here is passed as float8, which is stored as a foreign key in the facility table.

```
PREPARE route_nearest_facility (float8, float8, ,float8) AS
WITH nearest  node AS (
  SELECT id
  FROM pathway  edges  noded  vertices  pgr
  ORDER BY ST_Distance(the_geom, ST_SetSRID(ST_MakePoint($1, $2), 3857))
  LIMIT 1
),
event_nodes AS (
  SELECT id, name,
        (SELECT id FROM pathway_edges_noded_vertices_pgr ORDER BY
ST_Distance(the_geom, e.geom) LIMIT 1) AS node_id
  FROM facility e
  WHERE type_id=$3
),

paths AS (
  SELECT * FROM pgr_dijkstra(
    'SELECT id, source, target, cost FROM pathway_edges_noded',
    (SELECT id FROM nearest_node),
    ARRAY(SELECT node_id FROM event_nodes),  -- Pass multiple event nodes as an array
    false
  )
)
,
path_cost AS (
SELECT r.end_vid as node, SUM(r.cost) AS total_cost
FROM paths r
GROUP BY r.end_vid
ORDER BY r.end  vid
),

shortest_node AS (
    SELECT node as id
  FROM path_cost
  ORDER BY total  cost
  LIMIT 1
),
-- select * from shortest_node;

route AS (
  SELECT * FROM pgr_dijkstra(
    'SELECT id, source, target, cost FROM pathway_edges_noded',
    (SELECT id FROM nearest_node),
    (SELECT id FROM shortest_node),
    false
  )
)

SELECT e.*
FROM pathway_edges_noded e
JOIN route r ON e.id = r.edge;

EXECUTE route_nearest_facility(2075924.4, 7236775.2,2)
```

Info | Table | Preview | Query (city_festival_management)

Saved query | Name | Save | Delete | Load File | Save As File

```
52
53
54 EXECUTE route_nearest_facility(2075924.4, 7236775.2, 2);
55
```

Execute | 32 rows,0.013 seconds | Create a view | Clear | Query History

	id	old_id	sub_id	source	target	geom	cost	reverse_cost
1	425	178	1	331	169	0102000020...	10.68202751...	10.68202751...
2	183	12	6	169	170	0102000020...	5....	5....
3	184	12	7	170	140	0102000020...	15.02735816...	15.02735816...
4	147	8	10	139	140	0102000020...	18.18790083...	18.18790083...
5	146	8	8	138	139	0102000020...	5....	5....
6	145	8	7	137	138	0102000020...	14.31187923...	14.31187923...
7	144	8	6	136	137	0102000020...	9....	9....
8	143	8	5	135	136	0102000020...	9....	9....
9	142	8	4	134	135	0102000020...	10.48553501...	10.48553501...

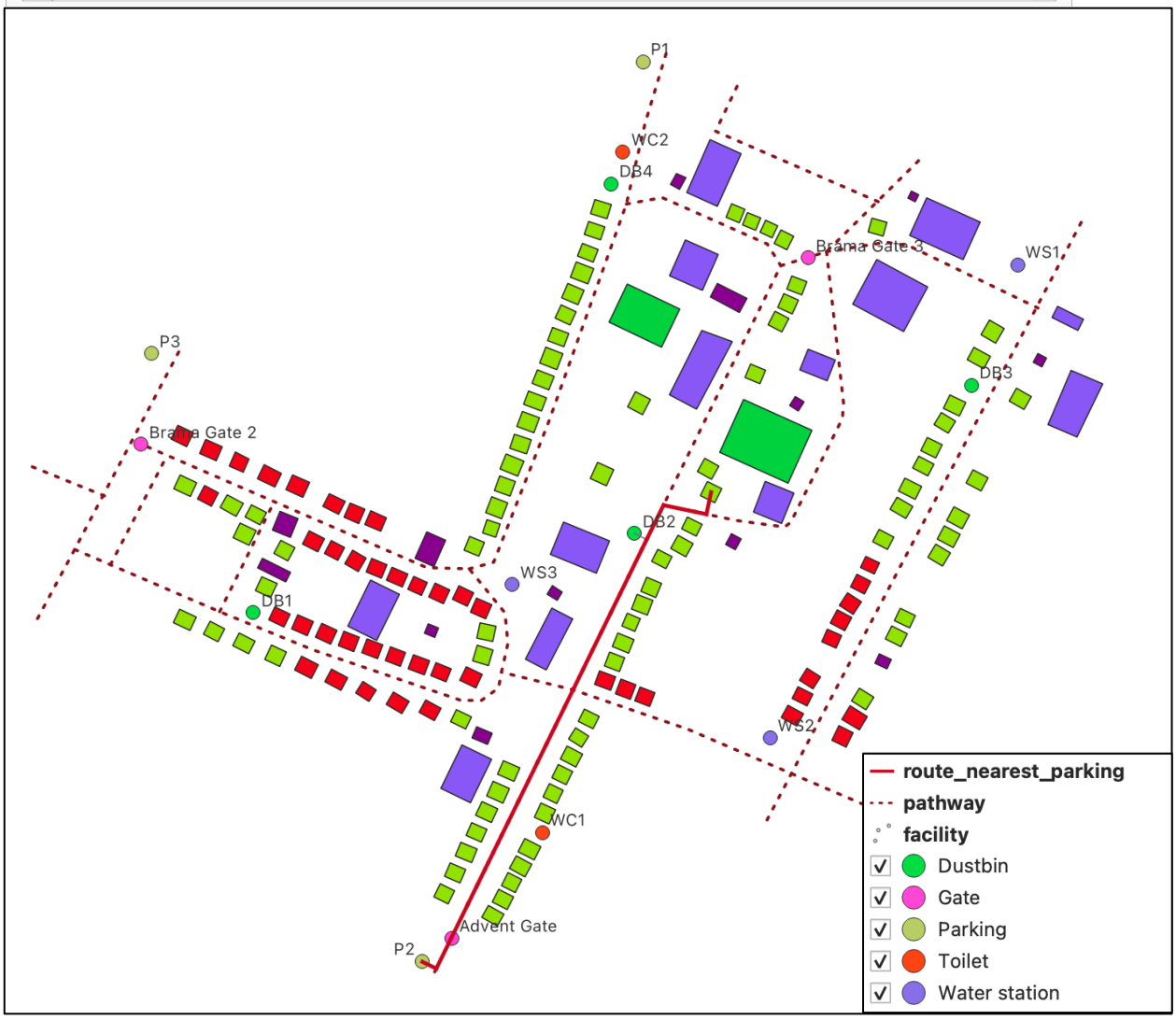


Figure 13: Route to the nearest parking facility

b. Route nearest attraction

Similarly, the above query can be updated to get the shortest path to the attractions.

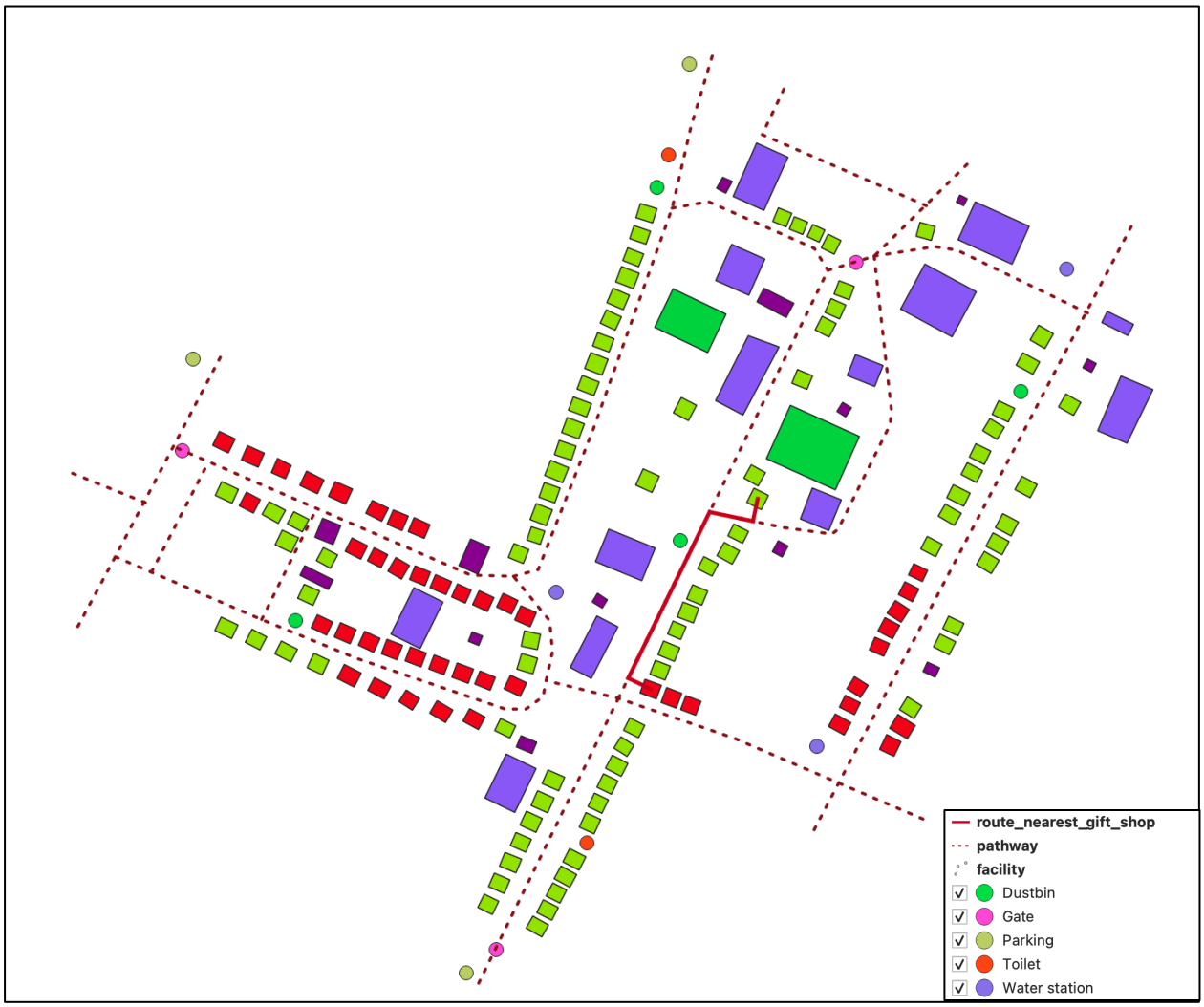


Figure 14: Route nearest gift shop

Roles and permissions need to be updated accordingly for the users to access these routing queries by the visitors, staff, and managers. In addition to this, as attraction or facility points might be added, changed, or removed, the tables related to routing need to be updated.

7 EXPORT DDL AND DML

The following pg_dump command is executed in a terminal to export DDL, DML, and both.

-s in command exports schemas (DDL) only, **-a** in command exports INSERT data (DML) only, **-d** exports both DDL and DML. By using these SQL, the database can be easily replicated.

```
pg_dump -U postgres -h localhost -p 5432 -s city_festival_management > DDL_city_festival_management.sql
```

```
pg_dump -U postgres -h localhost -p 5432 -a city_festival_management > DML_city_festival_management.sql
```

```
pg_dump -U postgres -h localhost -p 5432 -d city_festival_management > city_festival_management.sql
```

8 CONCLUSION

In this project, a spatial database was created and developed from scratch for the Gdańsk Christmas fair. With this, the broad concept of spatial data preparation, the importance of building the ER diagram, normalizing the table, denormalization, user roles and permission, and spatial queries based on user locations, non-spatial queries, and shortest path routing were fully exploited.

REFERENCES

The magic of Christmas in the center of Gdansk. Crowds at the Christmas Market. (2024, Dec 9). From TVN24: https://tvn24.pl/trojmiasto/jarmark-bozonarodzeniowy-w-gdansku-2024-atrakcje-produkty-smakolyki-st8186405